Practice Quiz

General Principles of Antimicrobial Therapy AND

Beta-Lactam Antibiotics, Other Cell Wall Inhibitors, and Membrane-Active Drugs, Part 1 and Part 2

Some items may have more than one answer.

The initial questions are intended to help students with pharmacology-related "must know" facts about pathogens.

To help you quickly find the answers for many of the questions, I have supplied page numbers of my Notes Handouts.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices.

Goals of Practice Quizzes:

1. To help students identify what you know and what you understand by applying your knowledge to clinical case vignettes.

2. To help students get used to thinking through the "Board-style" case vignettes and questions. NYITYCOM, COMLEX and USMLE examine students' medical knowledge using these types of questions.

HINT: Identify the information (clues) in the vignette that you need to answer the question. Don't be distracted by the information that is not needed to answer the question.

If you wish to use them, all answers can be found by doing a "search" of my Notes Handout and/or the textbook chapters designated in the resources of each question. The handout and

Drugs and Bugs – Mechanisms, Mechanisms, Mechanisms
The clinical use of antibiotics is to eradicate pathogenic microorganisms.
To select the right drug, the clinician must know the microbiology of the pathogens.
Students, you need to know the bugs (bacteria).
And, of course, the clinically relevant pharmacokinetic properties and PK-PD profile is necessary knowledge for optimization of dosing and patient care.

Textbook readings are not a required, but you may find one or the other to be VERY helpful

- **Pharmacology bridges physiology, pathophysiology, and clinical practice.**

When will the answers be posted?

THE ANSWERS WILL BE POSTED NOV 21 BEFORE THE PRECOMP EXAM.

Practice points and some microbiology points are included in the answers document to provide ***context and clinical relevance***. Students are not expected to memorize them for this course. My hope is that they will help you understand the pharmacology by conceptualizing it.

Resources are listed at the end of the document.

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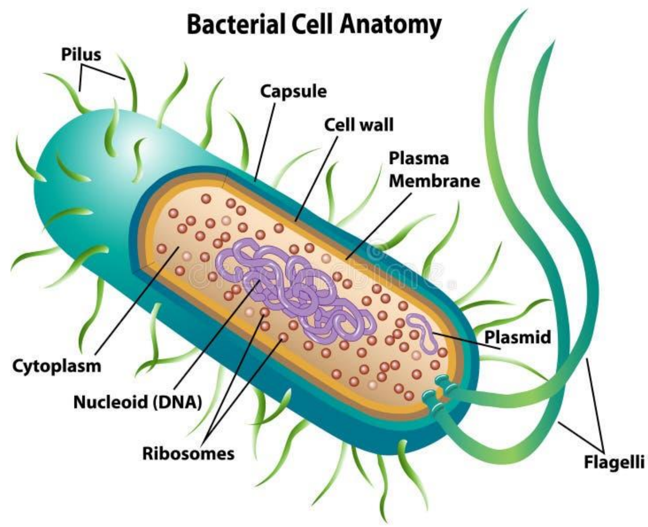
Updated September 2025

Learning Tip:
ELABORATION Finding additional layers of meaning by asking yourself "why" and "how" questions improves your ***mastery*** of new material and multiplies the mental cues available to you for later recall and application.



https://www.dreamstime.com/illustration/bacterial-capsule.html

PREFACE: Bacteria review relevant to antibiotic pharmacology

Relevance

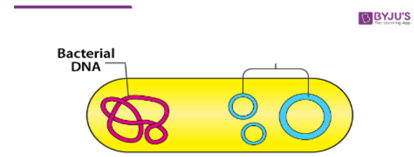
Knowing these very basic concepts will help you understand which drugs are used for which bacterial infections and why (mechanisms of action and resistance related to therapeutic uses), even if you haven't had microbiology yet.

a. What is the term for a microscopic single-celled organism that does not have specialized organelles or a nucleus with a nuclear membrane?
b. What is the term for bacteria having a spherical or oval shape?
c. What is the term for bacteria having a straight or curved shaped?
d. What bacteria have a rigid, thick peptidoglycan wall, teichoic acids covalently linked to N-acetylmuramic acid (NAM) monomers, and lack an outer membrane?
e. How are these bacteria categorized in terms of Gram stain?
f. What bacteria have a lipopolysaccharide outer layer, a thin peptidoglycan layer, and a periplasmic space?
g. What is the name of the proteins that allow hydrophilic solutes to diffuse across the outer layer of these bacteria?
h. How are these bacteria categorized in terms of Gram stain?
i. What bacteria are spiral shaped and either rigid or flexible and undulating?

https://www.britannica.com/science/

j. Some bacteria have a polysaccharide layer that protects them from the immune system and contributes to virulence. What is this layer called?

k. Many bacteria contain small, usually circular, covalently closed, double-stranded DNA separate from the chromosome ("extra-chromosomal"). What are these structures called?



https://byjus.com/biology/what-does-bacterial-plasmid-contains/

l. Why are these extrachromosomal structures important in bacteriology and crucial in pharmacology?

m. Some bacteria grow and maintain cellular functions if oxygen is available (metabolism by respiration) and in the absence of oxygen by fermentation. What is the term used to classify microorganisms able to use both metabolic pathways?

n. A large and diverse family of bacteria fitting this metabolism classification (above) are colonizers of the lower GI tract microbiome. What is the name of this large bacterial family? (Handout p2)

o. Why is knowledge of this order of bacterial crucially important in clinical practice?

p. Some bacteria can exist in a small, dehydrated, metabolically quiescent (quiet) state in response to limited nutrients or adverse conditions (survival morphotype). Then, with appropriate stimulation, germinate into single vegetative cells. What are the metabolically quiescent morphotypes called?

q. Which organism of the type described in item p. is an anaerobic gram-positive bacillus that is resistant to nearly all antibiotics and causes infection as a result of antibiotic therapy?

r. Some bacteria cause community-acquired pneumonia caused by organisms not visualized on a Gram stain or cultured using traditional techniques AND they are intrinsically resistant to beta-lactam antibiotics. What are they?

YOU CAN FIND THE ANSWERS IN the Notes Handout, Microbiology textbook, ScholarRx.

Answers:

a. Prokaryote: The cytoplasm contains a single double-stranded chromosome and is packed with ribosomes. This simple structure facilitates very rapid growth.

b. Coccus – cocci

c. Bacillus – bacilli – rods

d. Gram-positive cocci and bacilli

Cocci: *Streptococci, Staphylococci, Enterococci,* others

Rods: *Clostridium, Listeria, Corynebacterium, Bacillus anthracis, others*

> Peptidoglycan forms a rigid cell wall. It is a polymer of sugar molecules (NAM, NAG) and peptides crosslinked by transpeptidases.

e. Gram-positive: The purple iodine-dye complex (stain) is retained when treated with alcohol-acetone solvent.

f. Gram-negative cocci and bacilli

   Cocci: *Neisseria meningitidis, N. gonorrhoeae, Moraxella catarrhalis*

   Coccobacilli: *Haemophilus* spp, *Acinetobacter*, *Kingella*, *Francisella*

   Rods: Enterobacteriaceae, *Bacteroides*, *Pseudomonas,* others

g. Porins

h. Gram-negative: The purple iodine-dye complex (stain) is extracted when treated with alcohol-acetone solvent.

i. Spirochete

j. Capsule. If the capsule is amorphous, it is called a slime layer.

k. Plasmid

l. The structures are called plasmids. The genetic material in plasmids, particularly in gram-negative bacteria, can be transmitted to other bacteria and other bacterial species via hair-like tubular projections on the surface of the bacterial cell called sex pili (conjugative pili). The genes for resistance proteins, such a beta-lactamases, are often transmitted between bacteria, leading to antibiotic resistance among diverse species. Infection by gram-negative bacteria expressing Multidrug resistant beta-lactamases is a worldwide concern. These bacteria often carry genes for resistance to other classes of antibiotics.

   Other pili mediate virulence in some pathogenic gram-positive and gram-negative strains by functioning as adhesins. They enhance the pathogens' ability to attach to host tissues, allowing the bacteria to colonize surfaces and cells.

m. Facultative anaerobes: The majority of bacterial pathogens can utilize both respiration and fermentation metabolic pathways.

   Organisms that survive in the presence of oxygen have superoxide dismutase and catalase, which detoxify the superoxide radicals produced during respiration.

   Microaerophilic bacteria grow well in environments of 5%-10% oxygen.

n. Enterobacteriaceae: Gram-negative facultative anaerobic rods

   E. coli, Shigella, Klebsiella, Salmonella, Yersinia, Serratia, Citrobacter, to name a few pathogenic species

o. The Enterobacterales are constituents of the gut microbiome. Many of them contribute to health and are harmless. When outside the gut, they can cause infection. Multidrug resistance among Enterobacterales species is a worldwide problem.

p. Spores (endospores): A few gram-positive species produce spores. The medically important spore-formers are the gram-positive *Clostridium* spp. (tetanus, gas gangrene, *C. difficile*-associated colitis) and *Bacillus anthracis* (anthrax).

q. Clostridioides difficile

r. The so-called atypical bacteria that cause community-acquired "atypical" pneumonia are:

- *Mycoplasma pneumoniae*: Lacks a cell wall
- *Chlamydia pneumoniae*: Chlamydiae are obligate intracellular organisms and lack peptidoglycan. (C. psittaci also causes pneumonia. It is acquired from feral and domesticated birds and other animals.)
- *Legionella pneumophila*: Obligate intracellular bacteria present in phagocytes

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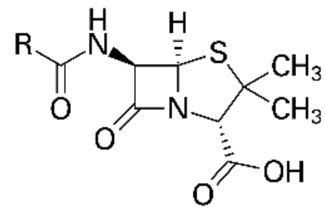
1. Thinking about the structural feature of the beta-lactam antibiotics necessary for its activity, explain how a particular enzyme expressed by bacteria, which is especially prevalent in gram-negative bacteria, inactivates the drugs.



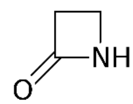
Short phrase

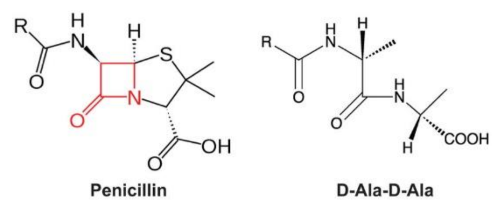
Hints: Relate it to the structure activity relationship.

Beta-Lactams Notes Handout, pages 6,8, and 10.

Brick: General Microbiology > Antimicrobial Agents > Penicillin
https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_MIC0013

Rationale: The beta-lactam ring is a structural analog of the D-alanyl-D-alanine on the pentapeptide chain of the peptidoglycans that compose the bacterial cell wall. The beta-lactam ring is essential for the activity of the drugs. Many bacteria express beta-lactamases that cleave the beta-lactam ring and inactivated the drugs.





Figures:

Upper left: Penicillin core structure by By Yikrazuul - Own work, Public Domain,
https://commons.wikimedia.org/w/index.php?curid=7348509

Upper right: Beta-lactam ring by Lhynard at English Wikipedia - Own work, Public Domain,

https://commons.wikimedia.org/w/index.php?curid=17454795

Lower: Penicillin and D-Ala-D-Ala https://www.researchgate.net/figure/The-mimicry-of-beta-lactam-antibiotics-to-D-alanyl-D-alanine-D-Ala-D-Ala-The_fig2_262043691

ScholarRx Brick: General Microbiology > Antimicrobial Agents > Penicillin https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_MIC0013

Learning Tip: **GENERATION**
The mind becomes more receptive to learning when you attempt to answer a question or solve a problem before being shown the answer or solution.

i) Where is this target located in the microbes?

A. Cell wall*
B. Cytoplasmic membrane
C. DNA
D. Ribosome

ii) What is the ultimate killing effect of this antibacterial action?

A. Apoptosis
B. Autophagy
C. Depolarization
D. Osmotic rupture*
E. Oxidative stress

Rationale: Cell wall / Osmotic rupture

Beta-lactams bind irreversibly to transpeptidases (penicillin binding proteins, PBPs), interfering with the function of the transpeptidases (PBPs), which are enzymes that catalyze the crosslinking of the peptide chains of the peptidoglycan chains.

This action triggers bacterial autolysins (the cell wall repair crew), which leads to osmotic rupture (self-destruction, basically) of the bacterial cells.

Nonlytic mechanisms probably also are involved.

Microbiology: Autolysins participate in the normal remodeling of the cell wall by breaking down small sections to allow for peptidoglycan biosynthesis.

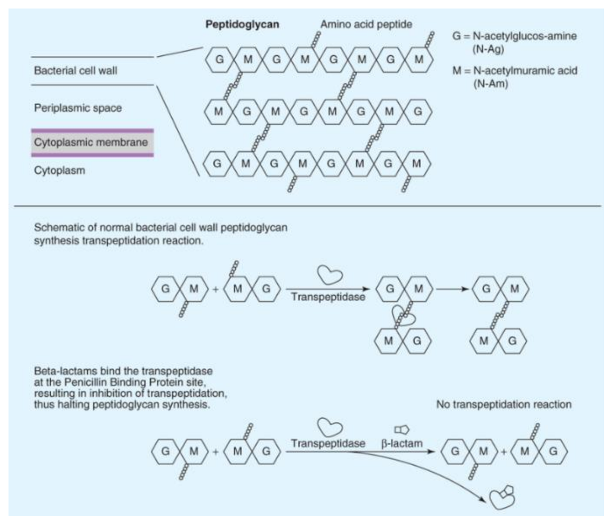


Figure: Access Medicine Katzung Basic & Clinical Pharmacology, 14e, 2018; Figure 43-5

3. A medical student shadowing a family physician is asked to explain what is meant by selectivity when used in reference to antimicrobial therapy. The student correctly responds:

A. The drug harms microbial cells but does not damage host cells.*
B. The drug targets only one type of microbial “receptor”.
C. The proportion of actual positives that are not overlooked (few false negatives).
D. The proportion of actual negatives that are correctly identified (few false positives).

Rationale: Antimicrobials that are toxic to the infective pathogen but do not harm host cells are said to have selective toxicity.

Incorrect options:

- The drug targets only one type of microbial "receptor": Not applicable. The term "selectivity" for certain receptors refers to drugs used in mammalian systems. Specificity is synonymous in this context.
- The proportion of actual positives that are not overlooked. ***Specificity*** measures the proportion of negatives that are correctly identified – the extent to which true negatives are classified as negative, such as the percentage of healthy people who are identified as not having some illness. A test has high specificity if it has a low false-positive rate.

  Example: It may not be feasible to use a test with low specificity for screening, since many people without disease will be positive, and potentially receive unnecessary diagnostic procedures.
- The proportion of actual negatives that are correctly identified (few false positives): ***Sensitivity*** of a test measures the positives that are correctly identified – the extent to which true positives are not overlooked, such as the percentage of sick people correctly identified as having illness in question. A test has high selectivity if it has a low false-negative rate.

New York State Department of Health Disease Screening – Statistics Teaching Tools

https://www.health.ny.gov/diseases/chronic/discreen.htm#:~:text=A%20highly%20specific%20test%20means,potentially%20receive%20unnecessary%20diagnostic%20procedures.

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4. An 8-year-old child is given oral penicillin VK every 8 hours for 10 days to be taken on an empty stomach with water for the treatment of streptococcal pharyngitis. Thinking about the class properties of the penicillins subclass, what is the antibiotic's route of elimination?

Short answer

Rationale: Except for the penicillinase-resistant penicillins (nafcillin, oxacillin, dicloxacillin, cloxacillin), the drugs in this class are eliminated renally as unchanged drug.

FYI: Penicillin VK is the potassium salt of penicillin V, which is more stable (bioavailability 65%) in stomach acid than penicillin G (poor oral bioavailability). Taking the tablet 1 hour before or 2 hours after eating enhances absorption.

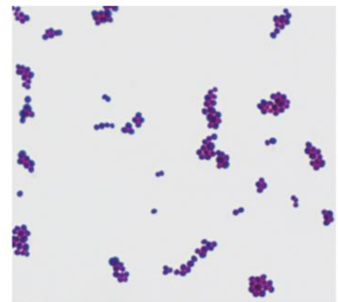
Learning Tip: **APPLICATION**
Applying your knowledge to practice – case vignettes at this point – help you assess your understanding.

5. K.C. is a 42-year-old healthy man presenting with a large, painful "boil" on the back of his neck. It started as a small, tender bumps "a few days ago" and has gotten progressively worse. He denies fever or chills. No known drug allergies (NKDA).

Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine, 8th Edition*: www.accessmedicine.com

The involved area is violaceous and indurated with multiple pustules draining externally around multiple hair follicles. Gram stain shows gram-positive spherical structures in grape-like irregular clusters.

He is given a diagnosis of carbuncles. The abscesses are drained; a sample of the exudate is sent for culture and susceptibility testing.



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

Gram-stain of Staphylococcus aureus showing Gram-positive cocci in pairs, tetrads, and clusters. Original magnification × 1000. (Courtesy of L. Ching.)

Citation: Chapter 13 The Staphylococci, Riedel S, Hobden JA, Miller S, Morse SA, Mietzner TA, Detrick B, Mitchell TG, Sakanari JA, Hotez P, Mejia R. Jawetz, *Melnick, & Adelberg's Medical Microbiology*, 28e; 2019. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=217770681&bookid=2629&Resultclick=2 Accessed: October 30, 2020

An e-prescription is submitted for an oral antibiotic with excellent activity against a penicillinase-producing species of the infective organism. He is directed to take it four times a day for 5 days one hour before eating or two hours afterward with at least four ounces of water and that it might cause nausea or heartburn, which is temporary and not harmful.

What antibiotic has been prescribed to this patient for definitive therapy?

A. Amoxicillin
B. Cefdinir
C. Dicloxacillin*
D. Penicillin VK
E. Vancomycin

ii) If K.C. reported that he got a rash from penicillin when he was a teenager, what drug would be an effective alternative?

Short answer

Definitions

Carbuncle, a superficial skin infection: Multiple abscesses developing in hair follicles, sebaceous glands or sweat glands that form a connected area of infection under the skin. Common areas are the back of the neck, face, axillae, and buttocks. It is caused by a commensal organism of skin flora, which also colonizes anterior nares.

Hint: The Gram stain findings are hallmark microscopic features of *S. aureus*.

Rationale: C. dicloxacillin

Culture and sensitivity identified methicillin-sensitive *S. aureus*, so the antibiotic was changed to a ***highly active*** agent directed against MSSA – definitive therapy:

i) **Dicloxacillin**

- **Preferred therapy:** The subclass called the "penicillinase-resistant" penicillins – nafcillin, oxacillin, dicloxacillin, and cloxacillin – are *highly active* against MSSA (methicillin-sensitive *S. aureus*) and are the preferred agents, especially for serious MSSA infections. Nafcillin and oxacillin have low oral bioavailability so are parenteral-only drugs.
- For superficial skin/soft tissue infections, such as furunculosis and carbuncles, caused by MSSA, dicloxacillin is preferred. It has moderate bioavailability and must be taken 4 times daily on an empty stomach with four ounces of water.

Must know facts about *Staphylococcus aureus*

- gram-positive cocci, facultative anaerobic
- grape-like irregular clusters on Gram stain
- catalase-positive
- coagulase-positive
- commensal bacterium of skin flora
- colonizes anterior nares
- penicillinase-producing (penicillin G/V are ineffective)
- resistance by beta-lactamase and altered PBP → PBP2a (mecA gene) = MRSA
- Empiric therapy should cover MRSA.

ii) **Cephalexin**: First-generation oral cephalosporin, cephalexin (oral) or cefazolin (IV, IM)

- Alternative option, especially in patients with penicillin allergy.
- Excellent activity against *Streptococci* spp and MSSA but is less resistant to penicillinases expressed by *Staphylococcus*.
- If a penicillinase-resistant penicillin cannot be used, other oral drugs effective for the treatment of MSSA are the *first-generation* cephalosporins, cefazolin and cephalexin.
- Regarding treatment of susceptible *Streptococcus* infections:
    - ❖ Penicillin G has excellent activity and a narrow spectrum. It has very low activity against gram-negative bacteria.

      Cefazolin and cephalexin also have excellent activity but have a moderate spectrum of activity – they will kill susceptible PEcK – *Proteus mirabilus, E. coli, and Klebsiella* – may affect the bacterial flora.
- **Beta-lactams are not effective against MRSA** (except ceftaroline, parenteral use only).
- **MRSA mechanism of resistance:** Target alteration – expression of PBPs that do not bind the drug (***PBP2a encoded by the mecA gene***).

➢ **Practice point, Selecting optimal therapy:**

Questions to ask yourself:

1. What drug ***allergies*** does the patient report?
2. What are the likely infective pathogens that will determine selection of empiric therapy?

3. Does the patient exhibit risk factors for resistant strains of the infective organism that should be covered by empiric therapy?
4. With these points in mind, what drugs are likely to be highly effective?
5. Among the choices, what is the preferred antibiotic therapy?

   Criteria: A drug or drug combination with:
   - Excellent activity against infective organism(s)
   - A favorable dosing schedule (adherence concern or nursing time concern)
   - Convenient administration
6. Will the therapeutic blood levels be achievable at the site of action with the selected formulation, route of administration, and dosing schedule and be adequate to eradicate the infectious organism?

Incorrect options:

- A/D. Amoxicillin and penicillin VK have low activity against penicillinase-producing *S. aureus*. Said another way, they are inactivated by *S. aureus* penicillinase.
- C. Cefdinir is an oral 3rd-generation cephalosporin that has activity against MSSA. However, it is an oral 3rd-generation cephalosporin with a broad spectrum of action – potential to disrupt the microbiome (commensals and pathogens) and cause secondary infections – *C. difficile* diarrhea and *Candida*.
- E. Vancomycin is IV only for the treatment of MRSA.

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6. A 27-year-old woman (G1P0) at 36 weeks gestation presents to the ED with fever, chills, myalgia, back pain, nausea, vomiting, and diarrhea. She reports that she was at a birthday party 10 days ago where she ate cheeses, deli meats, raw vegetable salad, and chocolate ice cream cake. She has had a healthy pregnancy to date. NKDA. Blood cultures are obtained, and the patient is admitted for intravenous antibiotic treatment of presumptive listeriosis.

The selected beta-lactam antibiotic, which was developed with an amino side chain to improve bacterial penetration and binding affinity, is highly active against the multidrug resistant (MDR), gram-positive rod *Listeria monocytogenes*. This drug is one of only a few that is effective against this organism.

Definition G1P0: Pregnant for the first time and has not yet delivered

G: gravida; P: para

i) Which one of the following best fits the description of the selected antibiotic?

A. Ampicillin*

B. Aztreonam

C. Cefazolin

D. Meropenem
E. Nafcillin
F. Vancomycin

ii) What is the structure-activity relationship of the amino side chain?

Hint: Beta-lactams Notes Handout p27 and p29.

Rationale: A. Ampicillin

Ampicillin is highly active against *L. monocytogenes*, and the preferred drug.

The clue in this question is that the antibiotic is, “an amino side chain”, which is intended to signal to the student that the selected drug is an aminopenicillin.

- **Practice point:** The symptoms are generally mild – flu-like symptoms, muscle aches, nausea, vomiting. However, the potential impact due to fetal *Listeria* infection can lead to fetal loss, premature labor, neonatal sepsis, meningitis, and death.

  The mild symptoms may delay diagnosis and treatment of listeriosis.

Structure-activity relationship: Addition of the amino side chain to the penicillin structure enhances the ability of these antibiotics to pass through porins to reach the cell wall transpeptidases of gram-negative bacteria AND improve binding affinity to PBPs.

However, amino side chain does not increase susceptibility to beta-lactamase producing strains, including penicillinase-producing *S. aureus*. Addition of the beta-lactamase inhibitor does restore the aminopenicillins’ activity.

The drugs in aminopenicillins group are ampicillin, amoxicillin, and piperacillin

Piperacillin is the antipseudomonal penicillin.

Microbiology: *L. monocytogenes* is an aerobic (facultative anaerobic), motile, beta-hemolytic, non-spore-forming, short gram-positive rod that exhibits characteristic tumbling motility by light microscopy. Listeria produces a characteristic appearance on blood agar with small zones of clear beta-hemolysis around each colony.

L. monocytogenes is widespread in the environment, especially in soil, decaying vegetable matter, and water.

Must know facts about *Listeria monocytogenes*
- gram-positive rod, facultative anaerobe,
- facultative intracellular
- beta-hemolytic
- catalase positive
- widespread in the environment
- can live in cold (including the refrigerator) or hot temperatures and salty environments
- foodborne transmission, dairy and meat

Listeriosis:
- gastroenteritis, self-limiting in immunocompetent person
- severe infection in the immunocompromised, pregnant, newborn, and elderly individuals: Listeriosis can lead to bacteremia, meningitis, encephalitis
- infection can be passed to the fetus → miscarriage, stillborn, neonatal sepsis
- **Few antibiotics are effective**. **Ampicillin is highly active; cephalosporins are ineffective.**

The bacteria can survive in soil for many months. They can live in a wide range of conditions and environments – acidic conditions, salty conditions, high temperatures, and low temperatures (such as refrigerators, frozen raw beef products, and ice cream).

Listeria can contaminate many foods, such as raw meat, hot dogs, deli meat, raw vegetables, raw and smoked seafood, fresh fruit, unpasteurized milk and cheese products.

FYI: Listeriosis:

- Listeriosis is usually a self-limited gastroenteritis (mainly diarrhea) in patients with intact immune systems.
- Listeriosis is an important bacterial pathogen in neonates, immunocompromised patients, older adults, pregnant patients, and occasionally, previously healthy adults.
    - These patients have a higher risk of severe listeriosis: **bacteremia, meningitis, encephalitis**
- Listeria has a predilection for placental and fetal tissue – the bacterium can cross the placenta and invade and survive in syncytiotrophoblasts, and infect the fetus → miscarriage, stillborn, neonatal sepsis.

➢ **Practice point:** In patients with serious infections (bacteremia/meningitis), or at risk of serious infection (above), ampicillin is highly active and the recommended antibiotic.

Videos: Two short videos you might want to view:

1) ***Tumbling motility of Listeria monocytogenes*** > Check out YouTube video

2) ***Listeria a unique model of infection biology*** by Cossart Lab Institut Pasteur > https://www.youtube.com/watch?v=dlAPOa_QXAo

Incorrect options:

- B. Aztreonam: Gram-negative aerobes only
- C. Cefazolin: All cephalosporins are INeffective against Listeria.
- D. Meropenem: Low activity against listeriosis plus very broad-spectrum
- E. Nafcillin: Ineffective against listeriosis
- F. Vancomycin: low activity

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7. An adult patient with right-sided endocarditis caused by VRE is given antibiotic therapy with a drug active against gram-positive bacteria only. It apparently kills bacteria by causing leakage of potassium and loss of membrane potential, resulting in rapid death by interrupting DNA, RNA, and protein synthesis. To mitigate an adverse effect associated with the drug, the patient's plasma creatine kinase (CK) levels are monitored.

i) What does VRE stand for?

ii) What is the name of the drug?

iii) What drug-induced adverse effect is signaled by the elevated plasma CK levels?

A. *C. difficile* superinfection
B. Hepatocyte damage
C. Myocardial infarction
D. Myopathy*
E. Renal dysfunction

Hint: Beta-lactams Notes Handout p61.

Rationale:

i) VRE is the acronym for vancomycin-resistant *Enterococcus*.

ii) Daptomycin

iii) Creatine kinase (CK) is the preferred muscle enzyme to measure in the evaluation of myopathies. Damage to muscle causes the creatine phosphokinase to leak from the muscle fiber into the blood. The CK-MM isoenzyme predominates in skeletal muscle, while creatine kinase-myocardial bound (CK-MB) is the marker for cardiac muscle.

- **Practice point:** Also monitor for signs/symptoms of additional daptomycin toxicities, which can lead to rare but serious adverse drug events: eosinophilic pneumonia, peripheral neuropathy, eosinophilia and systemic symptoms (DRESS), tubulointerstitial nephritis.

Must know facts about
Enterococcus faecalis and E. faecium

- gram-positive cocci, facultative anaerobic
- they do not produce toxins
- found in soil, water, plants, sewage, food, and on human skin, oral cavity, and large intestine
- adhere to catheters, dental prostheses, heart valves and form biofilms, causing persistent infections
- survive antiseptics and disinfectants and carried on health workers hands → hospital-acquired infections
- *E. faecium* is **intrinsically resistant to many antibiotics.** It expresses of PBPs with low affinity to beta-lactams. Some strains of *E. faecalis* have higher than usual MICs of penicillin and/or imipenem due to amino acid changes in PBPs. *E. faecalis* also expresses penicillinase. Some *E. faecium* isolates have been reported to have the β-lactamase gene.

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8. A 40-year-old female hematopoietic stem cell transplant recipient receiving bone marrow-ablative conditioning with chemotherapy is treated for febrile neutropenia with empiric antimicrobial therapy that includes an antibiotic effective against a broad range of gram-positive and gram-negative bacteria, including *Pseudomonas aeruginosa*. The selected drug is co-formulated with cilastatin. Patient monitoring includes signs and symptoms of seizures.

What class is this drug in?

A. Carbapenems*

B. Cephalosporins
C. Cyclic lipopeptides
D. Glycopeptides
E. Monobactams
F. Penicillins

Hint: Beta-lactams Notes Handout p51-52

Rationale: Carbapenems

Cilastatin is the only clue needed to get the right answer: Imipenem is the only antibiotic combined with cilastatin.

- The carbapenems are very broad-spectrum antibiotics. Activity of imipenem and meropenem includes *Pseudomonas aeruginosa*. The carbapenem without antipseudomonal activity is ertapenem.
- The drugs are stable in the presence of many beta-lactamases – an important feature of their broad spectrum of activity.

Important caveat: Carbapenems are inactivated by *Klebsiella pneumoniae* carbapenemases (KPCs).

What if cilastatin were not a clue?

- Clues: empiric therapy, broad range of activity (broad spectrum antibiotic) including *P. aeruginosa*.
- The antibiotics that fit the description are:
    - Piperacillin-tazobactam
    - 4^{th}-gen cephalosporin: Cefepime
    - Meropenem or imipenem-cilastatin (but not ertapenem)

CHALLENGE: For each of the three antibiotics above, think of a clue to be added to the stem that would point to each as the ONE BEST ANSWER.

- **Practice point, Neutropenia:** Febrile neutropenia is a **medical emergency**. Empiric broad-spectrum antibiotics should be started as soon as possible (within 60 minutes of triage) after blood cultures have been obtained, at full doses adjusted for renal function.

  The aim of empiric therapy is to cover the most likely and most virulent pathogens that may rapidly cause life-threatening infection in neutropenic patients.

  Causes: Cancer patients receiving:

  1) chemotherapy that suppresses the bone marrow function (myelopoiesis) and,

2) bone marrow ablative conditioning with chemotherapy in preparation for hematopoietic stem cell transplantation destroys hematopoietic cells in bone marrow, which produces profound pancytopenia.

The choice of antimicrobials (antibacterials and antifungals) is driven by multiple factors, including the degree of immunocompromise, prior antibiotic and infection history, local patterns of antibiotic resistance, and whether an agent is bactericidal or not.

Clinical information: UpToDate: Treatment of neutropenic fever syndromes in adults with hematologic malignancies and hematopoietic cell transplant recipients (high-risk patients), last updated May 3, 2023

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9. What is the function of cilastatin in the antibiotic co-formulation being given to the hematopoietic stem cell transplant recipient above?

Short phrase

Rationale: Cilastatin prevents inactivation of imipenem by dehydropepetidase-1.

Cilastatin blocks dehydropeptidase-1 in the brush boarder of proximal tubules, thus preventing inactivation by this enzyme. Imipenem-cilastatin achieves high concentrations in urine making it useful for complicated urinary tract infections, pyelonephritis and urinary tract infection with systemic signs and symptoms.

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10. AK is a 76-year-old male hospitalized for the treatment of hematogenous osteomyelitis resulting from hip replacement surgery. The area is surgically debrided and tissue samples are sent for culture and sensitivity testing. The infective organism is identified as MRSA. Therapy with a beta-lactam antibiotic is avoided.

What is this organism's mechanism of resistance to the entire beta-lactams group?

(exception: Ceftaroline and ceftobiprole, which are not currently used for treatment of osteomyelitis.)

A. Abnormally thick cell wall
B. D-alanyl-D-lactate expression
C. PBP with low affinity for beta-lactams*
D. Porins impenetrable to beta-lactams
E. TEM1 β-lactamases

Hint: Beta-lactams Notes Handout p9.

Rationale: *S. aureus* PBP2a encoded by the MecA gene (MRSA) is resistant to all beta-lactams except ceftaroline and ceftobiprole (the latter drug is not available in the US).

Incorrect options:

- A. Vancomycin resistant strains of *S. aureus* (VISA) have been found to develop an abnormally thick wall with increased numbers of D-Ala-D-Ala residues (“false targets”), which may sequester vancomycin within the cell wall; the underlying mechanism is not known.
- D. *S. aureus* is a gram-positive organism and therefore does not have an outer membrane and porins.
- B. Expression of D-alanyl-D-lactate instead of D-Ala-D-Ala by the VanA gene acquired from enterococci is a mechanism of *S. aureus* resistance to vancomycin. (This is a rare mechanism of *S. aureus* resistance.)

  E. Beta-lactams regain activity against Ambler class A,TEM-1-expressing beta-lactamases when a beta-lactamase inhibitor is added. However, MRSA resistance is due to target alteration, not hydrolyzing enzymes. Therefore, beta-lactams are not effective even with the addition of a beta-lactamase inhibitor.

Learning Tips:
RETRIEVAL PRACTICE
Quizzing is a reliable measure of what you have learned and what you have not mastered.
SPACED PRACTICE

11. What anti-MRSA drug from among the antibiotics studied in this block should be included in AK’s empiric treatment of osteomyelitis?

One word answer

Rationale: Vancomycin

Thinking it through:

What drugs are active against MRSA?

Vancomycin, daptomycin, ceftaroline

What are the recommendations regarding each of these drugs?

- Vancomycin is the antibiotic of choice for the treatment osteomyelitis due to MRSA – indeed, any presumptive MRSA infection for hospitalized patients (IV only administration) as long as resistance is not suspected.
  - Daptomycin is an alternative for vancomycin-resistant strains MRSA (VRSA) and vancomycin intermediate-resistant strains (VISA).

Note: Changing to an expensive alternative (daptomycin) without clinical advantage is not appropriate.

- Ceftaroline labeled and off-label uses are skins/skin structure infections (including MRSA), bloodstream infections directed against MRSA, and community-acauired, hospital-acquired, and ventilator-associated pneumonia caused by *S. pneumoniae*, MSSA, and some GNBs (gram-negative bacteria).

  It is not currently recommended for osteomyelitis treatment. Further studies are needed to determine ceftaroline's efficacy and safety in the treatment of osteomyelitis.

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12. A few minutes into the second infusion of the anti-MRSA drug, AK complains of itching. He denies dyspnea and pain. Flushing and erythema are observed on his neck, face, and trunk. His vital signs are within normal limits. It is noted that the rate of infusion on the infusion pump is set too high.

What is the most likely cause of the characteristic symptoms and signs of the patient's drug reaction?

A. Anaphylaxis
B. Non-febrile elevation of body temperature
C. Non-immunogenic histamine release*
D. Prostaglandin D2 release
E. Sympathetic nervous system stimulation

Hint: Beta-lactams Notes Handout p57

Rationale: Vancomycin can directly act on mast cells and induce histamine release. The syndrome is called non-immune anaphylactoid infusion-related reaction (red man syndrome).

Prevention: Increasing the infusion time and/or further diluting vancomycin in a larger quantity of diluent (commonly saline solution).

Comparison to allergy/anaphylaxis, an immunogenic reaction: Hives and itching; flushed or pale skin; hypotension; swollen tongue or throat and bronchoconstriction, which can cause wheezing and trouble breathing; a weak and rapid pulse; nausea, vomiting or diarrhea; dizziness or fainting

Other vancomycin adverse effects:

- IM injection: necrosis
- Nephrotoxicity
- Ototoxicity

The risk of nephrotoxicity and ototoxicity increases with high doses and concurrent use of other drugs causing same adverse effects.

---

13. Which one of the following best represents the currently recommended vancomycin dosing strategy for the treatment of serious MRSA infections? (more than one answer)

   A. Clinical efficacy correlates with high peak dose to MIC ratio that produces a greater microbial kill rate.
   B. Clinical efficacy correlates with the total amount of drug in the body determined by the AUC/MIC.*
   C. Clinical efficacy correlates with drug concentrations above the MIC for the large majority of the time.*
   D. Clinical efficacy correlates with high doses that shift the IC50 on the dose-response curve to the right.

Hint: Notes Handout Beta-lactams p54

Rationale: Clinical efficacy correlates with total amount of drug in 24 h period.

T>MIC with a short organism-specific post-antibiotic effect
(Vancomycin's PK-PD profile was recently redefined.)

Clinical efficacy correlates with total amount of drug in the body – 24h AUC/MIC ratio:

The ideal vancomycin dosing regimen maximizes the amount of drug received in a 24-hour period and may reduce the risk of nephrotoxicity.

This dosing strategy may decrease the nephrotoxicity risk of vancomycin.

- Monitor renal function
- **Practice point:** For drugs with AUC-MIC PK-PD profiles, the ratio must be at least a certain value to correlate with efficacy for a given drug for a given bug (microbe).

  This number is determined by outcomes-based evidence.

Example: For vancomycin in the treatment of serious MRSA infections, the AUC/MIC should be >400.

---

14. i) What is vancomycin's antibacterial target? (What does vancomycin bind to?)

ii) What step in bacterial cell wall synthesis is inhibited?

   A. Formation of short peptide interbridges
   B. Inhibition the bactoprenol carrier molecule

C. Lipidation of lipopolysaccharide membrane

D. Polymerization of peptidoglycan monomers*

Hints: Beta-lactams Notes Handout p55

Rationale:

Vancomycin inhibits cell wall synthesis in gram-positive bacteria by tightly binding the D-alanyl-D-alanine terminus of the pentapeptide chain, which inhibits the enzyme transglycosylase – the penultimate step in cell wall synthesis. Further elongation of the peptidoglycan and cross-linking are prevented.

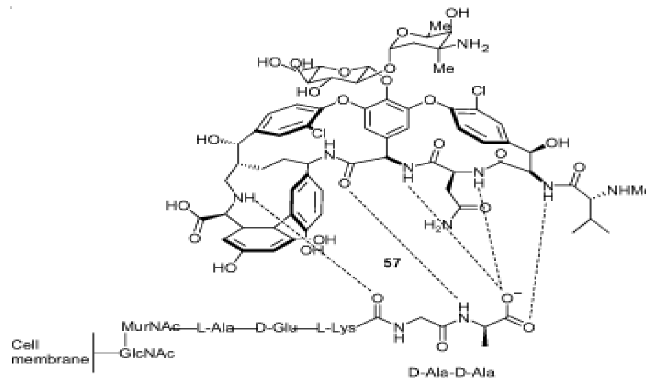

Figure 23. Binding of vancomycin to the D-Ala-D-Ala terminus of a peptidoglycan precursor.

15. From a Big Picture perspective, what is a way to think about the generations of cephalosporins that might make them easier to remember?

Hint: Think about the drugs in each generation according to their spectrums of activity against aerobic and facultative anaerobic gram-positive and gram-negative bacteria – in the ideal (nonexistent) world where there is no resistance.

	Gram-positive	Gram-negative
1st-gen.		
2nd-gen.		
3rd-gen.		
4th-gen.		
Advanced-gen.		

Answers: The five generations (subclasses) are

	Gram-positive	Gram-negative Enterobacterales
1st-gen. Cefazolin (IM, IV) Cephalexin (oral)	Streptococci Staphylococci including penicillinase-producing and oxacillin-resistant strains	PEcK Proteus mirbilus, E. coli, Klebsiella pneumoniae
2nd-gen. Cefuroxime (IM, IV, oral)	As for 1st-gen. (1st-gen. are more active)	PEcK plus the respiratory pathogens *Haemophilus influenzae* and *Moraxella catarrhalis*

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2^{nd}-gen. cephamycins: Cefoxitin (IM, IV)	As for cefuroxime	PEcK, a few more Enterobacterales, and *Haemophilus* Gram-negative anaerobes including Bacteroides fragilis
3^{rd}-gen. Ceftriaxone Cefotaxime	Streptococci Penicillin-resistant *S. pneumoniae* MSSA Good CSF concentrations	Extended- spectrum of activity against Enterobacterales *Haemophilus* *Neisseria meningitidis and N. gonorrhoeae* Spirochetes: *B. burgdorferi* (Lyme disease), *Leptospira*, *Treponema* (penicillin preferred)
3^{rd}-gen. anti-*Pseudomonas* Ceftazidime, Ceftazidime-avibactam Ceftolozane-tazobactam	Low activity compared to ceftriaxone	Enterobacterales *Pseudomonas aeruginosa* Poor activity against anaerobes, including *B. fragilis*
4^{th}-gen. Cefepime	Streptococci including penicillin-resistant *S. pneumoniae* Staphylococci (MSSA)	Enterobacterales Respiratory pathogens *Neisseria meningitidis*
Advanced-gen. Ceftaroline	MSSA, MRSA Penicillin-resistant *S. pneumoniae*	PEcK *Haemophilus, Moraxella*
Advanced-gen. Cefiderocol	None	Multidrug-resistant, aerobic, ESBL-producing and AmpC-producing gram-negative bacteria

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16. A 43-year-old woman is being prepared for a laparoscopic hysterectomy. To reduce the risk of surgical wound infection from mixed enteric gram-negative bacilli, including the obligate anaerobic gram-negative rod, *Bacteroides fragilis*, she is given one prophylactic I.V. infusion of an antibiotic 30 minutes prior to the surgery. In what subclass is this antibiotic categorized?

A. Aminopenicillins
B. Carbapenems
C. Cell membrane-active drugs

D. Cephamycins*
E. Glycopeptides

Information to help you deduce the correct option:

Requirements of antimicrobials used for prophylaxis: The selected agent should be:

- bactericidal,
- nontoxic,
- inexpensive, and
- should have in vitro activity against common organisms that cause wound infection after a specific surgical procedure.

Broad spectrum agents should be avoided.

Hint: Summary Table on p47 of Beta-lactams Notes Handout

Rationale: Cephamycins, a subclass of the 2nd-generation cephalosporins

Antimicrobial prophylaxis should be used in circumstances in which ***efficacy has been demonstrated and benefits outweigh the risks*** of prophylaxis for prevention of wound infection in contaminated procedures.

➢ **Practice point:** Infusion of the first antimicrobial dose should begin within 60 minutes before surgical incision and the drug should be discontinued within 24 hours of the end of surgery. The short duration decreases the potential toxicity, microbial resistance, and cost.

Cephamycins unique feature: Activity against gram-negative obligate anaerobes, including *B. fragilis* and other *Bacteroides* strains.

➢ **Practice point:** Because cefoxitin and cefotetan are active against the common gram-negative aerobic and anaerobic bacilli, including many strains of *Bacteroides*, and have stability to many plasmid-mediated beta-lactamases, the cephamycins can be used to treat mixed anaerobic infections, such as peritonitis, diverticulitis, and pelvic inflammatory disease. *B. fragilis* resistance to cefoxitin and cefotetan is increasing.

Must know facts about *Bacteroides fragilis*

- gram-negative rod, obligate anaerobe
- minor component of intestinal microbiome
- polysaccharide capsule stimulates abscess formation
- leading cause of intra-abdominal abscesses
- **Broad resistance:** MDR GNB (multidrug-resistant gram-negative bacillus)
- produces beta-lactamases (unusual for anaerobes) that inactivate penicillins and cephalosporins, except the cephamycins.
- Beta-lactamase inhibitors inactivate the beta-lactamases expressed by *B. fragilis*.

Incorrect options:

– The aminopenicillin-beta-lactamase inhibitor, ampicillin-sulbactam, is active against enteric pathogens, including *B. fragilis*, but it is a broad-spectrum antibiotic.

– The carbapenems are highly active against enteric pathogens, including *B. fragilis*, but they are broad-spectrum antibiotics – the broadest spectrum of the beta-lactams.
– The cell membrane-active drug, daptomycin, is active against gram-positive organisms only.
– The glycopeptide, vancomycin, is active against gram-positive organisms only.

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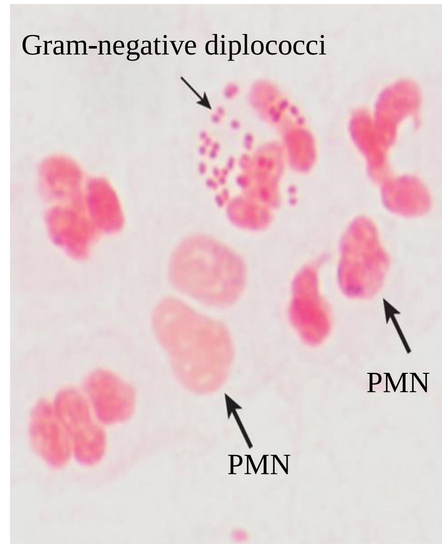
17. A 22-year-old sexually active male presents to the ED with complaint of painful urination and presence of a profuse yellow discharge at the urethral meatus of 4 days duration. No known drug allergies (NKDA). Gram-stain of urethral exudate shows gram-negative diplococci in many polymorphonuclear (PMN) leukocytes. The patient is given a diagnosis of uncomplicated gonococcal urethritis (gonorrhea). The standard of care for the treatment this infection is administered. What is it and what dose?



A. Amoxicillin
B. Aztreonam
C. Ceftriaxone*
D. Cefuroxime
E. Ciprofloxacin
F. Doxycycline
G. Penicillin G benzathine

I believe the dosing recommendations for uncomplicated gonorrhea will be useful knowledge in your clinical rotations.

Rationale: Ceftriaxone has the lowest rates of gonococcal resistance in epidemiologic surveys.

I will not ask you to remember specific doses on the exams and the Level 1 and Step 1 board exams will not ask you questions about specific doses.

Important Practice Update
Updated CDC Guidance (Dec 18, 2020)
Ceftriaxone 500 mg I.M. x 1 dose as monotherapy for uncomplicated gonorrhea. The 500 mg dose is for patients weigh less than 300 pounds. For patients >300 pounds, 1 g I.M. x1 dose is recommended.
If chlamydia cannot be ruled out through microbiologic testing, doxycycline 100 mg orally twice daily for 7 days is the updated guidance.

Azithromycin orally is no longer recommended "as a strategy for preventing ceftriaxone resistance and treating possible coinfection with *Chlamydia trachomatis*."

The concerns: "Increasing concern for antimicrobial stewardship and the potential impact of dual therapy on commensal organisms and concurrent pathogens, in conjunctions with the low incidence of ceftriaxone resistance and the increased incidence of azithromycin resistance."

Resources: Update on the CDC's Treatment Guidelines for Gonococcal Infection, 2020; MMWR Dec 18, 2020 / 69(50);1911–1916 https://www.cdc.gov/mmwr/volumes/69/wr/mm6950a6.htm

Access Medicine Jawetz, Adelberg, and Melnick's Medical Microbiology, 28e, 2019; Chapter 20: The Neisseriae; Figure 20-1

Legend: Gram stain of urethral exudate of a patient with gonorrhea. Nuclei of many polymorphonuclear cells are seen (*large arrows*). Intracellular Gram-negative diplococci (*N. gonorrhoeae*) in one polymorphonuclear cell are marked by the *small arrow*

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18. Hauser's *Antibiotic Basics for Clinicians* lists the "Six Ps" mnemonic of bacterial resistance mechanisms to the beta-lactam antibiotics: Penicillinase, PBPs, Porins, Pumps, Penetration, and Peptidoglycan. Explain each mechanism.

Hint: Beta-lactams Notes Handout p8.

Rationale: The 6 Ps:

- Penicillinases / Beta-lactamases – the most common mechanism: Bacteria that express enzymes beta-lactamases that hydrolyze the beta-lactam ring and inactivate the drugs are resistant. Penicillinases specifically inactivate penicillins.
- PBPs: Bacteria with mutational alterations of the genes expressing transpeptidases (PBPs) that do not bind the drugs will be resistant. Also, bacteria that naturally express PBPs that do not bind the drugs are intrinsically resistant. This is a target alteration mechanism.
- Porins: Beta-lactams that cannot pass through (permeate) the porins of gram-negative bacteria will not reach their target – the transpeptidases of the cell wall. Again, the drug does not reach the target.
- Pumps: Gram-negative bacteria that express efflux pumps in their outer membrane will push out the drugs. Therefore, again, the drugs will not reach their targets of action. Gram-positive organisms do not have an outer membrane.
- Penetration: The beta-lactams do not penetrate cell membranes into the cytoplasm. Therefore, obligate intracellular organisms, which reside in host cells, are not susceptible to the actions of beta-lactams. Therefore, the drug does not reach the target.

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- Peptidoglycan: Organisms that lack peptidoglycan in their cell walls will not be susceptible to the beta-lactams.

The 6 Ps fit into the three of the four general mechanisms of resistance, which are:

(1) Drug does not reach target

(2) Drug inactivation

(3) Target alteration

(4) Organism expresses alternative metabolic pathways (not a mechanisms of beta-lactam resistance)

Resources: LWW Hauser's Antibiotic Basics for Clinicians, 2e, 2013

Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. Mcmichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition

Cellulitis with ill -defined erythema on the lower leg. Note the subtle, streaky lymphangiitis extending proximally.

Citation: Chapter 151 Cellulitis and Erysipelas, Kang S, Amagai M, Bruckner AL, Enk AH, Margolis DJ, McMichael AJ, Orringer JS. *Fitzpatrick's Dermatology, 9e*, 2019, Figure 151-1. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=210430491&bookid=2570&Resultclick=2 Accessed: October 15, 2021

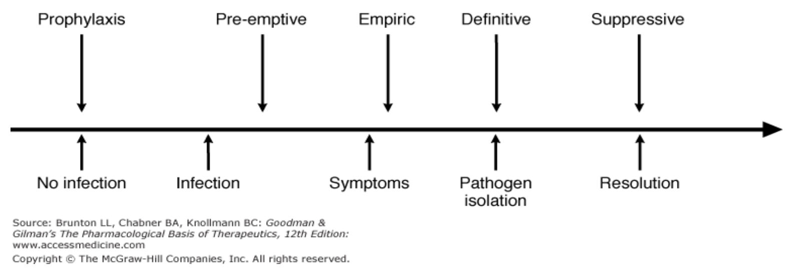
19. Mr. Nelson is a 60-year-old man who presents to the PCP with a painful, erythematous, nonpurulent infection on his right lower extremity. It began as a small "sore" 3 days ago and rapidly progressed. Possible penicillin allergy manifested by a generalized urticarial rash on his upper body, but no angioedema or shortness of breath, when he took penicillin VK for a "throat infection" as a teenager.

All vital signs are within normal range. Exam reveals an edematous lesion with ill-defined erythema that is warm to the touch and streaky lymphangitis extending proximally. The patient is given a clinical diagnosis of cellulitis and an e-prescription is submitted for an oral antibiotic effective against the presumed infective pathogens, beta-hemolytic streptococci (*S. pyogenes)* and *Staphylococcus aureus*.

Which of the following options best represents this approach to therapy?

A. Definitive therapy
B. Empiric therapy*
C. Preemptive therapy
D. Prophylactic therapy
E. Suppressive therapy

Rationale: Empiric therapy



Administration of antibiotic therapy before the infective pathogen is known is recommended when a delay in initiating antibiotic therapy could cause significant morbidity or fatality. Empiric therapy is also called presumptive therapy.

- **Practice point:** Cellulitis is a common infection of the deep dermis and subcutaneous tissues, most often caused by bacteria. It presents with the classic signs of inflammation: redness (rubor), heat (calor), swelling (tumor), and pain (dolor). Unilateral lower extremity involvement is typical and systemic symptoms are usually absent. Cellulitis is treated with an antibiotic. Recurrences are common.
- **Immunology, Penicillin allergy:** Penicillins – all beta-lactams – are not proteins. The drugs or their derivatives become immunogenic by forming haptens with host proteins, such as red blood cells.

Integrating immunologic and pharmacologic concepts:

– Haptens are small molecules that elicit an immune response only when attached to a carrier protein, forming a hapten-carrier complex.

– Penicillins form a hapten-carrier complex with red blood cells. Immune attack of the drug-haptenated RBCs by IgG and activation of the complement system can lead to hemolytic anemia.

– Cross-reactivity can occur between beta-lactams with closely related structures. The side chains and ring structures are potentially immunogenic.

Cross-reactivity takeaway: Beta-lactams with identical side chains are more likely to cause allergic cross-reactivity than drugs with dissimilar side chains.

- **Clinical management, Expert recommendations:** Perform skin testing in patients who report penicillin allergy. If skin testing results are negative, then the patient is not a substantial risk of an immediate (type I) reaction to penicillin and may safely receive a cephalosporin or carbapenem (if not allergic to the specific drug administered).

If penicillin skin testing is positive, then there is a high likelihood the patient is allergic to penicillin, and 2-3% of these patients may react to a cephalosporin. Skin testing, or a test dose, of the cephalosporin or carbapenem may be warranted.

Incorrect options:

- Definitive therapy: When the microorganism and its susceptibility to specific antibiotics are identified, the initial empiric antibiotic therapy is changed to an effective agent with a narrower spectrum of activity targeting the specific microbe. The drug is given for the shortest duration of time necessary to eradicate the infectious pathogen.
- Preemptive therapy: Early targeted therapy in high-risk patients who already have a laboratory or other test indicating that an asymptomatic patient has become infected. For example, antibiotic treatment of an individual who has been in close contact with a patient with meningococcal meningitis (caused by *N. meningitidis*).
- Prophylactic therapy: An antimicrobial is administered to prevent infection in patients at high-risk, such as surgery to prevent a surgical infection.
- Suppressive therapy: Antimicrobial therapy given to a patient who has been treated but the original infection is not completely eradicated by the initial therapy, as in patients with an immunological defect. The goal is to achieve functional status, not necessarily a cure.

Fitzpatrick Dermatology, 9e, 2019; Chapter 151: Cellulitis and Erysipelas: Definition and Figure 151-1

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20. Mr. Nelson's skin testing is positive for penicillin allergy and negative for other antibiotics. His mild non-purulent cellulitis is treated empirically with an oral cell wall synthesis inhibitor that has reliable activity against the potential infectious microbes, *Streptococcus* spp and penicillinase-producing *S. aureus*, and has a relatively narrow spectrum of activity. It is unlikely to trigger an allergic reaction in this patient.

The patient has no risk factors for MRSA.

i) Which one of the following best fits this description of the selected drug?

A. Dicloxacillin
B. A first-generation cephalosporin*
C. A second-generation cephalosporin
D. A third-generation cephalosporin
E. A glycopeptide

ii) What is the name of the selected drug?

Rationale: A first-generation oral cephalosporin, cephalexin

Empiric therapy should be directed against *Streptococcus* spp and MSSA (penicillinase-producing *S. aureus*).

- **Practice point:** For infections where *Streptococcus* is established as the causative infectious agent (such as erysipelas), penicillin G is preferred because it is the most active against susceptible gram-positive bacteria.

  The patient reports a rash without angioedema or bronchospasm during previous penicillin therapy. Sensitivity testing is positive for allergy to penicillin but negative for allergy to cephalosporins. Cross-reactivity with cephalexin is low.

Incorrect options:

– Natural penicillins, penicillin G, is highly active against *Streptococcus* spp but ineffective against penicillinase-producing *S. aureus*.

– Second-generation cephalosporins are less active against gram-positive bacteria than the first-generation cephalosporins and the penicillins class of antibiotics but are still effective. They are good options for patients with penicillin allergy.

– Third-generation cephalosporins, such as ceftriaxone, are extended-spectrum drugs. A narrow spectrum drug is preferred for definitive therapy.

– A glycopeptide, such as vancomycin, is usually reserved for MRSA treatment, rather than MSSA. It kills staphylococci relatively slowly so, for MSSA, the beta-lactams mentioned above are preferred. Vancomycin is effective in the treatment of *Streptococcus* infections in penicillin-allergic patients.

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21. A usually healthy, active 17-year-old female presents to the pediatrician with complaint of a sore, scratchy throat, productive cough, stuffy, runny nose with thick, yellowish mucus, sneezing, headache, and general ill-feeling (malaise). The symptoms came on yesterday. Ibuprofen helped relieve the headache. NKDA.

Significant findings are T 99.9°F; pharyngeal erythema, nasal mucosal swelling and congestion; lungs are clear to auscultation. The patient is advised on the management of her illness.

With antibiotic stewardship practices in mind, which one of the following recommendations would be the most appropriate course of action for this patient?

A. Prescribe empiric amoxicillin
B. Prescribe empiric anti-influenza medication
C. Recommend ENT consult
D. Recommend symptomatic treatment*

Rationale: The patient is presenting with signs and symptoms of the common cold, which is a minor viral upper respiratory illness. Antibiotic therapy is not warranted. (Signs and symptoms are inconsistent with strep throat.)

https://www.cdc.gov/antibiotic-use/hcp/core-elements/outpatient-antibiotic-stewardship.html

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22. Amoxicillin's pharmacokinetic properties are generally typical of the beta-lactams class, in addition to some unique features.

i) Which choice, A or B, best describes amoxicillin's PK properties?

A	B
Oral	Oral
Excellent absorption	Bioavailability 50% to 75%, reduced by food
Excellent, reliable blood levels	Widely distributed
Widely distributed	Protein binding >95%
Low protein binding	Some hepatic metabolism, biliary excretion
Not metabolized	Excreted in feces / urine as unchanged and metabolites
Excreted in urine as unchanged drug	Dose adjustment in renal impairment not usually required.
2 to 3 times daily dosing	4 times daily dosing

ii) Which drug has PK properties represented in the other choice?

iii) Reflection: What is the clinical relevance of the drugs' pharmacokinetic profiles?

Rationale:

i) Choice A: Beta-lactams are widely distributed in body fluids and tissues, with significant amounts appearing in liver, lung, muscle, middle ear effusions, maxillary sinus secretions, bone, gallbladder, bile, muscle, ascitic and synovial fluids, but poor penetration of tissues with tight junctions – cerebral spinal fluid (blood-brain barrier), prostate, and vitreous humor of the eye (blood-retinal barrier). They are excreted in the urine as unchanged drug.

Amoxicillin's excellent oral absorption and reliable blood levels allow 2 to 3 times daily dosing. It's oral suspension is pleasant tasting – an important aspect in treating children.

ii) From among the beta-lactams, Choice B is representative of the pharmacokinetic properties of the oral penicillinase-resistant (antistaphylococcal) penicillin, dicloxacillin. The other drugs in the group available in the US – nafcillin and oxacillin – are parenteral only.

iii) Understanding and employing pharmacokinetic processes (ADME) and their interconnections ***increases the probability of therapeutic success and reduces the occurrence of adverse drug events***.

Pharmacokinetic processes link a drug's dose to its concentration in biologic fluids and, for most drugs, determines the intensity (concentration at the site of action) and time course of drug action. Pharmacokinetics provides the scientific basis for dose regimen design and dose selection.

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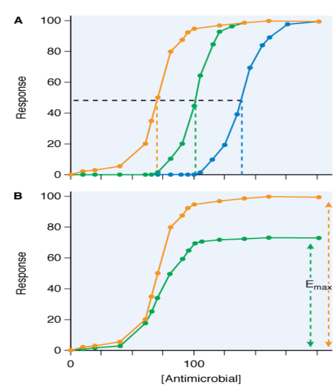
23. A 2-year-old boy is brought to the pediatrician's office by his mother. He has been tugging on his ear since yesterday after his morning nap and has been running a temperature of 101.3F (38.5C) taken by ear thermometer at home. He has been irritable; his appetite and fluid intake have been good. A bulging, white tympanic membrane is visualized on otoscopy. The patient has had two bouts of acute otitis media treated with antibiotics, 3 months ago and 4 weeks ago, respectively. He is given a clinical diagnosis of recurrent acute otitis media. An e-prescription is submitted for an antibiotic at a high dose intended to eradicate bacterial infection by *H. influenzae*, *M. catarrhalis*, and *Streptococcus pneumoniae* strains expressing intermediate resistance. Which one of the following statements best defines the profile of the gram-positive bacterial strain with the resistance pattern described in the vignette?

A. IC50 shifts to the right; Emax is achieved.*
B. IC50 shifts to the right; the response curve is depressed.
C. High peak concentration to MIC ratio (C_{max}/MIC) will achieve a greater rate of bacterial killing.
D. High dose increases the drug's half-life for improved efficacy.



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com

Rationale: Susceptibility of a microbe to a given antimicrobial agent shifts the sigmoid E_{max} curve:

Intermediate resistance: Shift to the right → ↑ IC_{50}. Susceptibility tests indicate that the minimum inhibitory concentration is increased. Therefore, higher antimicrobial dose is needed the achieve the higher plasma concentration that will achieve the desired effect – eradication of the microbe.

Incorrect options:

– B. The IC50 shifts to the right; the response curve is depressed = **Resistance:**

1- Decrease in E_{max} means that increasing the dose of the antimicrobial agent beyond a certain point will not increase the response of the microbe to the antimicrobial.

2- The increase in IC_{50} (right shift) of a particular antibiotic becomes so large that it is not possible to achieve a drug concentration adequate for killing the microbes without causing toxicity to the patient.

- C. The PK-PD profiles of the individual antibiotics are intended to predict dosing strategies related to clinical effectiveness. For example, PK-PD profile of the aminoglycosides is C_{max}/MIC model, the higher the dose, the greater the rate and effect of bacterial killing. These drugs also show prolonged persistent effects – continued suppression of bacterial growth after drug concentration decreases below the MIC.
- D. Increasing the dose (a pharmacokinetics strategy) does not increase the drug's half-life, which is a function of clearance. A high dose allows a longer time above the MIC, which, in turn, may allow reduced frequency of dosing – for example, twice daily instead of three time daily.

Resource: Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics 13e, 2018, Chapter 52: General Principles of Antibiotic Therapy; Goodman & Gilman Figure 52-3

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24. A 64-year-old man with a history of advanced hepatic cirrhosis and ascites presents to the emergency department with severe abdominal pain. He has had multiple visits to the hospital in the past few months, including multiple visits for ascites drainage and antibiotics. He has been taking fluoroquinolone antibiotic prophylactic therapy. His abdomen is distended with significant tenderness on palpation. Upon further evaluation, he is diagnosed with spontaneous bacterial peritonitis (SBP) severe disease. Ascitic fluid, blood, and urine samples are obtained for identification and susceptibilities; ascitic fluid is sent for PMN (polymorphonuclear leukocytes) count.

i) Which of the following is the best course of action?

A. Hold treatment while waiting for culture and sensitivity results.
B. Run DNA-based PCR testing for microbial resistance.
C. Initiate empiric broad-spectrum antibiotic therapy.*
D. Start the standard initial drug regimen for BPS (non-critically ill patients).

ii) What is the next step regarding antibiotic therapy that should be included in the treatment plan?

Rationale:

i) Empiric therapy with a broad-spectrum agent should be begun immediately after obtaining peritoneal fluid to maximize the patient's chance of survival.

Most cases of SBP are due to gram-negative bacteria of the gut microbiome and possibly *Streptococcus* and *Staphylococcus*.

ii) Definitive therapy: De-escalation to definitive therapy when culture and susceptibility test results are available.

Incorrect options:

- Holding treatment while waiting for culture and sensitivity results could lead to the patient's mortality.
- Running DNA-based PCR testing for microbial resistance might identify resistance genes but it does not provide information about the susceptibility of the organisms to specific antibacterial drugs. It is very expensive with no added benefit.
- The infection could be resistant to the standard initial drug regimen for BPS. The stem states that the patient has had multiple visits for drainage of the ascitic fluid and has received multiple antibiotics.

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25. A 19-year-old male presents to the ED with complaint of a lesion on his penis. He admits to having sex with a female sex worker about 1 month ago. He reports that he has never done anything like it before and has not had a sexual encounter since that time. He is otherwise healthy. He does not use recreational drugs, alcohol, or nicotine-containing products. NKDA.

Physical findings: a painless, non-suppurative chancre the penis, firm bilateral lymphadenopathy. All other findings are unremarkable. Serological testing confirms *Treponema pallidum* infection. Screening for HIV and other STIs is negative. He is given the standard treatment, which induces rapid resolution and long-acting effect. A follow-up appointment is scheduled with the hospital's family practice clinic.

What is the preferred drug for this patient's early-stage disease?

A. Amoxicillin 3 gm plus probenecid 500 mg both orally twice daily x14 days
B. Azithromycin 2 gm po x1 dose
C. Ceftriaxone 1 gm I.M. daily x10-14 days
D. Doxycycline 100 mg orally every 12 hours x 14 days
E. Penicillin G benzathine 2.4 million units, I.M. x1 dose*
F. Tetracycline 500 mg orally four times daily x14 days

Rationale: Penicillin G is the preferred antibiotic. Benzathine penicillin is a formulation to provide delayed and sustained absorption (depot formulation), resulting in prolonged blood and tissue concentrations.

All the other options are alternatives.

- Azithromycin is only a ***last resort*** due to treatment failures associated with macrolide resistance.
- The long therapy durations are because of *T. pallidum's* slow multiplication rate.

Caution: Adherence may be reduced with longer courses of therapy.

➢ **Practice point:** Knowing the alternatives, or knowing where to find the information, is important for managing treatment in penicillin-allergic patients and during periods of temporary drug shortages.

Selection of antibiotic treatment for this patient:

– Allergies: No known drug allergies. Penicillin may be given.

– Infection: *T. pallidum*, a spirochete

– Penicillin G is highly active against *T. pallidum* producing rapid resolution of infection, has a favorable dosing schedule (one dose for the long-acting benzathine formulation) providing sustained therapeutic blood levels, and a convenient (but painful) I.M. injection

– *T. pallidum* resistance to penicillin G has not been documented to date.

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26. A 2-week-old male infant is brought to the ED by his parents because of fever, irritability, lethargy, and poor tone for 2 hours. The patient is ultimately diagnosed with bacterial meningitis and admitted to the NICU for treatment. The combination antibiotic regimen includes cefotaxime. Another 3rd-generation cephalosporin with the same spectrum of activity is not used in this patient because of potential adverse effects: 1) displacement of bilirubin on plasma proteins, increasing the risk of kernicterus (encephalopathy), and 2) potentially fatal lung and kidney damage due to drug-calcium precipitates when calcium-containing fluids are coadministered.

i) Which one of the following is the described drug that should be avoided in this patient?

A. Cefazolin
B. Cefotaxime
C. Ceftazidime
D. Ceftriaxone*
E. Cefepime

ii) Which is the preferred drug therapy for this patient?

A. Cefazolin
B. Cefotaxime*
C. Ceftazidime

D. Ceftriaxone
E. Cefepime

Rationale: Although ceftriaxone would be effective in the treatment of meningitis in the neonate, cefotaxime, which has the same spectrum of action as ceftriaxone, does not displace bilirubin from plasma protein binding sites or complex with calcium when calcium-containing fluids are coadministered.

- **Practice point:** Cefotaxime is frequently in shortage due to manufacturing issues.
  What are alternatives to cefotaxime: If cefotaxime is not available, ceftazidime of cefepime are recommended alternatives against ampicillin-resistant gram-negative enteric bacteria and pneumococci (*S. pneumoniae*).

  The practicing clinician should always be aware of alternative therapies.

  Cefazolin is a 1st-generation cephalosporin with limited gram-negative activity and is inappropriate for the empiric treatment of meningitis.

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27. What properties of ceftriaxone and cefotaxime are the same and what properties are different?

Why is it important to think of these two drugs together?

CHALLENGE: Write down on paper as much as you can from memory. Then check. Then write it down from memory again.

Hint: Beta-lactams Notes Handout, p40

Rationale: Ceftriaxone and cefotaxime can be used interchangeably for various infections. Thinking of them together strengthens knowledge of their interconnection and may reduce the memorization burden.

More important, if one or the other is not available due to drug shortages (all too common), you have knowledge about a reasonable alternative.

	Different	
Same	Ceftriaxone	Cefotaxime
· Parenteral administration only · Widely distributed in the body	· **Not for use in neonates** · Highly plasma protein bound and can displace	· Safe in neonates Does not displace bilirubin from albumin binding sites

<table>
<tr>
<td>· CSF levels better than other beta-lactams; best CSF penetration when meninges are inflamed</td>
<td rowspan="2">unconjugated bilirubin from binding sites
· Risk of kernicterus
· Precipitation with Ca2+</td>
<td rowspan="2">Does not precipitate with Ca2+</td>
</tr>
<tr>
<td rowspan="2">· Spectrum of action – broad gram-negative activity (not Pseudomonas) and pneumococci
· Activity against susceptible organisms
· Empiric treatment of a wide range of serious infections
Treatment of patients of all ages (one caveat – see ceftriaxone neonates)
· They are both oxyimino-cephalosporins inactivated by ESBLs</td>
</tr>
<tr>
<td>· Longer half-life
Dosing 1x daily
(2x daily for serious infections)
· Active drug eliminated through the biliary system in feces and in urine</td>
<td>· Shorter half-life
Dosing every 8 hours
· Active drug and active metabolite eliminated in urine</td>
</tr>
</table>

ESBL: Extended-spectrum beta-lactamase

---

28. Empiric antibiotic therapy for meningitis in the premature neonate should be a bactericidal drug having coverage against Group B *Streptococcus*, *E. coli*, and *Listeria monocytogenes*.

i) Which one of the following is included for the eradication of the presumptive *Listeria* infection?

A. Ampicillin*
B. Cefazolin
C. Ceftazidime
D. Cefotaxime
E. Piperacillin-tazobactam

ii) Why is a bactericidal antibiotic preferred in the treatment of meningitis?

Rationale: Ampicillin is highly active against *L. monocytogenes.*

- *L. monocytogenes* is ***intrinsically resistant*** to cephalosporins, carbapenems, and aztreonam (gram-negative only spectrum). Vancomycin and daptomycin are also ineffective.

Pharmacologic properties of drugs for the treatment of meningitis:

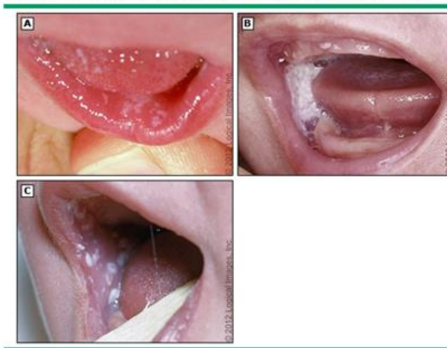
1) Bactericidal agents effective against the infective pathogen are preferred.

- The CSF is a site of impaired humoral immunity.
- Use of a bactericidal agent results in optimal cure and survival (based on studies in animals with pneumococcal meningitis).
- Clinical observations of poor outcomes in patients receiving bacteriostatic therapy.

2) High-dose antimicrobial therapy penetrates the CSF in therapeutic concentrations during meningeal inflammation. Although beta-lactams, including ampicillin, generally have low penetration into the CSF, meningeal inflammation causes separation of the endothelial tight junctions (blood-brain barrier), enhancing drug concentrations in the CSF.

3) The pharmacokinetics-pharmacodynamics profile (and evidence of efficacy) determines does and dosing frequency. The beta-lactams must maintain concentrations above the MIC: they exhibit a T>MIC PK-PD profile (this is also called concentration-independent profile) with minimal post-antibiotic effect.

Three examples of thrush (oral candidiasis) in an infant



Note the white plaques on the inner lip and tongue (panel A) and on the buccal mucosa (panels B and C).

29. A normally healthy, active 4-month-old female presents to the pediatrician with a white coating on her tongue and a diaper rash for 2 days. She has also been having loose bowel movements for one week. Medical history is pertinent for recurrent acute otitis media treated with oral amoxicillin-clavulanate for the past 8 days of a 10-days course of therapy.

Otoscopy shows that the acute otitis media has resolved. White plaques are noted on the buccal mucosa, palate, tongue, and oropharynx. Confluent erythema with erythematous papules and plaques is noted in the inguinal area. Findings are consistent with a drug-induced secondary infection. The exam is otherwise unremarkable.

The child’s mother is directed to continue the antibiotic until it is finished. She is given instructions on the use of a medication for management of the orpharyngeal infection and a topical cream for the diaper dermatitis. What secondary infection are these medications intended to treat?

Short answer

Candidal diaper dermatitis

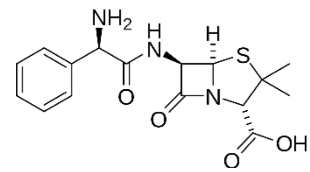
Erythematous plaques and satellite pustules in an infant.

Leslie Goldstein, Pharm.D.
Updated September 2025

Rationale: The correct answer options are:

Fungal infection / *Candida albicans* infection /candidiasis.

Antibiotic killing of susceptible commensal organisms of the regional microbiome allows overgrowth of organisms not susceptible to (not killed by) the antibiotic therapy. As a result, these organisms proliferate causing secondary infections, such as candidiasis, as in this case, or *C. difficile* disease.



- **Pharmacology point, Structure-activity relationship**: The aminopenicillins, ampicillin, amoxicillin, and piperacillin, are called *extended-spectrum penicillins* because the amino group enhances uptake of the drugs through bacterial porins, which increases their spectrum of activity to many enteric gram-negative organisms as compared to penicillin G.

UpToDate 2020 Graphic 69465 v. 6.0, Oropharyngeal thrush; Graphic59361 v. 9.0, Candidal diaper dermatitis (accessed from Candida infections in children)

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30. An adult with recurrent rhinosinusitis with indication for antibiotic therapy is given amoxicillin-clavulanate for coverage of ESBL-producing *Haemophilus influenzae* and *Moraxella catarrhalis*.

Which one of the following is a common classification of these beta-lactamases?

A. Ambler Class A serine beta-lactamases*
B. Ambler Class B metallo-beta-lactamases
C. Ambler Class C serine beta-lactamases
D. Ambler Class D serine beta-lactamases

What does ESBL stand for?

Rationale: Ambler Class A beta-lactamases; ESBLs: extended-spectrum beta-lactamases expressed in the presence of extended-spectrum cephalosporins, ceftriaxone, cefotaxime, ceftazidime, cefepime, and cefuroxime.

An important mechanism of antimicrobial resistance among many gram-negative pathogens is the plasmid-mediated expression of beta-lactamases.

ESBLs derived from TEM type, SHV type, and CTX-M family confer multidrug resistance to penicillins, cephalosporins, and aztreonam. The first-generation beta-lactamase inhibitors, clavulanic acid, sulbactam, and tazobactam, have activity against many, but not all, Class A

Leslie Goldstein, Pharm.D.
Updated September 2025

plasmid-mediated beta-lactamases expressed by Enterobacterales and other GNBs (gram-negative bacteria).

FYI: *H. Influenzae* and *M. catarrhalis* are members of the respiratory microbiome.

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31. A 78-year-old obese woman with a long history of type 2 diabetes is hospitalized for treatment of an acute complicated urinary tract infection. She resides in a nursing home and has been hospitalized three times in the past six months for the treatment of recurrent urinary tract infections. She is given intravenous infusions of meropenem-vaborbactam.

What bacterial enzymes are inhibited by the beta-lactamase inhibitor paired with the antibiotic in the patient's therapy?

A. Class A ESBLs*
B. Class A KPCs*
C. Class B metallo-beta-lactamases
D. Class C Amp C beta-lactamases*
E. Class D OXA beta-lactamases

Rationale: Class A ESBLs and KPCs and the AmpC beta-lactamases

Uropathogens that cause complicated UTIs are *E. coli* (most common), *Klebsiella* spp, *Proteus* spp, *Pseudomonas*, enterocci, MSSA, and MRSA.

Increasing rates of ESBL resistance in uropathogens is reported globally and KPC from clinical isolates of *K. pneumoniae* are increasingly isolated.

AmpC (Class C) beta-lactamases are typically encoded on the chromosome, but may be carried on plasmids, of many gram-negative bacteria, often grouped in the acronym SPACE *Serratia*, *Pseudomonas* or *Proteus*, *Acinetobacter*, *Citrobacter*, and *Enterobacter*.

	Class A	Class A ESBLs and KPC carbapenemases	Class D	Class C Ambler-type (AmpC)
Clavulanic acid	√			
Sulbactam	√			
Tazobactam	√			
Avibactam	√	√ (some)	√ (some)	√
Vaborbactam	√	√ (some)		√
Relebactam	√	√ (some)		√

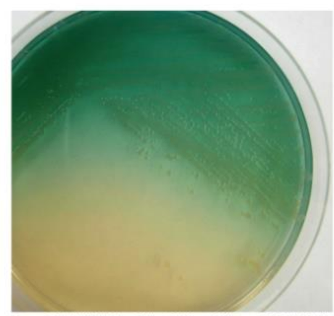
32. A 40-year-old woman is treated for a skin abscess with debridement and an oral drug with activity against streptococci and MRSA. Culture and sensitivity results show that the infection is caused by methicillin-sensitive *Staphylococcus aureus* (MSSA). The antibiotic regimen is changed to cephalexin. This course of action is referred to as:

A. Definitive therapy*
B. Preemptive therapy
C. Presumptive therapy
D. Suppressive therapy

Rationale: Also called directed therapy, definitive therapy is relatively narrow spectrum antimicrobial directed against the specific pathogen.

When the microorganism and its susceptibility to specific antibiotics are identified, the initial empiric antibiotic therapy is changed to an effective agent with a narrower spectrum of activity targeting the specific microbe. The drug is given for the shortest duration of time necessary to eradicate the infectious pathogen.

The goal is to optimize clinical outcomes while minimizing unintended consequences of antimicrobial use, including toxicity, selection of pathogenic organisms, secondary infections, and emergence of antimicrobial resistance.



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

P. aeruginosa on a 10-cm Mueller-Hinton agar plate. Individual colonies are 3–4 mm in diameter. The organism produces pyocyanin, which is blue, and pyoverdine, which is green. Together these pigments produce the blue-green color that is seen in the agar around the pseudomonas growth. (Courtesy of S Lowe.)

Citation: Chapter 16 Pseudomonas, Acinetobacter, Burkholderia, and Stenotrophomonas. Riedel S, Hobden JA, Miller S, Morse SA, Mietzner TA, Detrick B, Mitchell TG, Sakanari JA, Hotez P, Mejia R. *Jawetz, Melnick, & Adelberg's Medical Microbiology*, 28e; 2019. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=217771138&bookid=2629&Resultclick=2 Accessed: October 31, 2020

33. A 14-year-old female cystic fibrosis patient takes an inhaled monobactam antibiotic via nebulizer for the treatment of her chronic lung disease. The drug is being used for the management of infection caused by an obligate aerobic gram-negative, non-fermenting, oxidase-positive rod commonly present in moist environments. This organism colonizes mucous membranes or skin and invades locally. It can cause serious, potentially life-threatening infections in immunocompromised and medically vulnerable patients.

The selected drug has activity against aerobic gram-negative organisms only.

What is the selected antibiotic?

A. Aztreonam*
B. Cefazolin
C. Daptomycin
D. Ertapenem

Hint: Students, *you can come up with the correct answer from the last statement*. You do not need to know the microbiology of the organism.

Blue-green pigment visible on nutrient agar is unique to *P. aeruginosa*.

Notes p51

Rationale: Aztreonam

Clues:

1) Aztreonam is the only antibiotic (in the US) with the monobactam structure.

2) The spectrum of action is narrow – aerobic gram-negative rods, only

3) The drug is given by oral inhalation for the management of chronic lung disease in patients with cystic fibrosis.

Incorrect options:

- Cefazolin has activity against gram-positive bacteria and a few enteric gram-negative rods. It is not effective against *P. aeruginosa* and it is not administered by inhalation.
- Daptomycin is a cell membrane-active drug for the treatment of gram-positive infections not responsive to other antibiotics. It is administered by I.V. infusion.
- Ertapenem is a broad-spectrum carbapenem, but it is not active against *P. aeruginosa*. It is administered parenterally.

Resources: Jawetz, Adelberg, and Melnick's Medical Microbiology, 28e, 2019; Chapter 16: *Pseudomonas, Acinetobacter, Burkholderia*, and *Stenotrophomonas*; Figure 16-2

Legend: P. aeruginosa on a 10-cm Mueller-Hinton agar plate. Individual colonies are 3–4 mm in diameter. The organism produces pyocyanin, which is blue, and pyoverdine, which is green. Together these pigments produce the blue-green color that is seen in the agar around the pseudomonas growth. (Courtesy of S Lowe.)

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34. A 84-year-old mechanically ventilated male with a long history of chronic obstructive pulmonary disease develops new-onset fever, purulent sputum, leukocytosis, and sustained respiratory deterioration. Microscopic examination of bronchoscopic specimens shows gram-negative rods. Culture and sensitivity results show carbapenem-resistant Enterobacterales (CREs). The patient is given IV infusion with a siderophore antibiotic effective against multidrug resistant GNBs including some Ambler class D OXA-type beta-lactamases.

i) What is the unique property of this drug?

ii) What is the mechanism of antibacterial action of this drug?

iii) What is the name of the drug?

Rationale: Cefiderocol

i) Complexes with extracellular ferric iron and is actively transported across the outer membrane of aerobic gram-negative bacteria using iron transport systems. Cefiderocol also passes through porins by passive diffusion.

ii) Cefiderocol is a beta-lactam antibiotic. It binds and inhibits PBPs, interrupting cell wall synthesis.

Cefiderocol is stable to many beta-lactamases.

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35. Emily is a D.O. who starts her own family practice clinic in a rural area. She recognizes the need for a coordinated program to achieve optimal outcomes for the clinic's patients. One of her focus areas is the appropriate use of antimicrobial agents.

- She communicates with her clinical and office staff about best practices for antimicrobial use, including education of the patients and caregivers.
- She provides her clinical staff with current references and guidelines for effective antimicrobial selection.
- The office manager is assigned the tasks of displaying patient information pamphlets describing appropriate antibiotic use and printing daily reports on antibiotic prescriptions obtained from the EMR.
- Emily provides regular feedback to the individual prescribers and progress reports to the clinical and administrative staff.

What is this systematic measurement and coordinated interventions to promote the optimal use of antimicrobials is known as?

Short answer

Rationale: Antibiotic stewardship.

Antimicrobial stewardship involves coordinated programs in doctors' offices, hospitals and other health care facilities, and on the farm. They:

– Promote the appropriate use of antibiotics and other antimicrobials
– Improve patient outcomes
– Reduce microbial resistance, and
– Decrease the spread of infections caused by multidrug-resistant organisms.

http://www.apic.org/Professional-Practice/Practice-Resources/Antimicrobial-Stewardship

Goals of Antibiotic Stewardship Programs, To:

1. Reduce antibiotic consumption and inappropriate use
2. Reduce *Clostridiodes difficile* infections
3. Improve patient outcomes
4. Increase adherence/utilization of treatment guidelines
5. Reduce adverse drug events
6. Decrease or limit antibiotic resistance
   - Hardest to show
   - Best data for health-care associated gram negative organisms

Tamma PD, Cosgrove SE. Infect Dis Clin North Am. 2011 25:245; Ohl CA, Luther VP. J. Hosp. Med. 2011;6:S4

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36. Matching: Sometimes it is easier to remember the few drugs active against a certain microbe than memorizing the many that are not.

P. aeruginosa can cause serious and potentially life-threatening infections in immunocompromised and medically vulnerable patients.

P. aeruginosa is intrinsically resistant to many antibiotics, can form biofilms on tissues and medical devices, produces enzymes and toxins damaging to host tissues, and can lead to sepsis.

Highlight the beta-lactams that have activity to susceptible *P. aeruginosa*. (In other words, P aeruginosa is not intrinsically resistant to them.)

(Only commonly used parenteral agents are listed.)

Penicillins	**Cephalosporins**		**Carbapenems**	**Monobactam**
Penicillin G	1st-gen	Cefazolin	Imipenem-cilastatin	Aztreonam
Ampicillin	2nd-gen	Cefoxitin	Meropenem	
Piperacillin-tazobactam	3rd-gen	Ceftazidime	Ertapenem	
	3rd-gen	Ceftriaxone		
	4th-gen	Cefepime		
	Adv-gen	Ceftaroline		
	Adv-gen	Ceftolozane-tazobactam		
	Novel	Cefiderocol		

Why is it important to know about infections caused by *P. aeruginosa* and the agents used to treat them?

Answers:

Penicillins	**Cephalosporins**		**Carbapenems**	**Monobactam**
Penicillin G	1st-gen	Cefazolin	Imipenem-cilastatin	Aztreonam
Ampicillin	2nd-gen	Cefoxitin	Meropenem	
Piperacillin-tazobactam	3rd-gen	Ceftazidime	Ertapenem	
	3rd-gen	Ceftriaxone	Doripenem	
	4th-gen	Cefepime		
	5th-gen	Ceftaroline		
	5th-gen	Ceftolozane-tazobactam		
	Novel	Cefiderocol		

***P. aeruginosa* infections have a high mortality rate.**

- *P. aeruginosa* is an obligate aerobic gram-negative organism common in the environment, especially water (including hot tubs). It can contaminate respiratory equipment and sinks.
- It tends to infect immunocompromised hosts and burn patients.
- It is the most serious pathogen and the most common gram-negative pathogen causing ventilator-associated pneumonia.
- Multidrug resistant *P. aeruginosa* strains have become common in many hospitals and regions.

➢ Remember: If an organism is intrinsically resistant to an antibiotic, adding a beta-lactamase inhibitor will not increase the drug's effectiveness, unless the intrinsic resistance is due to a beta-lactamase that would be inhibited by the inhibitor.

P. aeruginosa resistance mechanisms:

Resistance mechanisms:

– ESBLs

– AmpC beta-lactamases

– Carbapenemases

– Efflux pumps

Porin channel mutationsResources: The Sanford Guide to Antimicrobial Therapy, 50e, 2020; Table 4A: Antibacterial Activity Spectra

UpToDate: Epidemiology, microbiology, and pathogenesis of Pseudomonas aeruginosa infection, last updated Oct 19, 2021; literature review current through Oct 2021.

Infection and Osteopathic Principles and Practice:

What changes in a patient's lymphatic tissues would be expected in response to infection?

LWW Health Libraries, Medical Education; An Osteopathic Approach to Diagnosis and Treatment, 4e, 2021; Chapter 107: Lymphatics > Infection

https://meded-lwwhealthlibrary-com.arktos.nyit.edu/content.aspx?bookId=2969&sectionId=248816511&resultClick=1#248816548

Answer

In response to infection, the lymphatic tissue reacts much in the way a factory might to increased demand. Production of immunocompetent cells and destruction of the invading organisms or substances are increased. Lymphoid tissue that was previously nonpalpable may hypertrophy. Upper respiratory infections frequently have findings of lymphadenopathy in the cervical chain lymph nodes. Inguinal lymph node adenopathy may be noted in cases of urinary or pelvic infection. Enlargement of the spleen, known as *splenomegaly*, occurs during diseases such as mononucleosis. Because it is normally covered by the stomach and the costal cartilage, a palpable spleen is a cardinal finding of this viral infection.

The spleen may also become enlarged when the marrow is unable to further undertake hematopoiesis. The primitive production of fetal hemoglobin can be reestablished and may help maintain the blood's oxygen-carrying capacity in spite of idiopathic anemia. The liver may increase in size because of infection or other toxic influence. Hepatitis, or inflammation of the liver, frequently results in *hepatomegaly*. Subsequent to infection, it may be quite some time before the lymphatic tissue regresses to its previous size, if at all.

Resources

Resource: Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics 13e, 2018, Chapter 52: General Principles of Antibiotic Therapy

Access Medicine Basic & Clinical Biostatistics, 5e, 2020; Chapter 12: Methods of Evidence-Based Medicine and Decision AnalysisAccess Medicine Katzung & Vanderah's Basic & Clinical Pharmacology; 15e, 2021; Chapter 43: Beta-Lactams and Other Cell Wall- & Membrane-Active Agents Access Medicine Sherris & Ryan's Medical Microbiology, 8e, 2022; Chapter 21: Bacteria – Basic Concepts > Bacterial Structure https://accessmedicine.mhmedical.com.idm.iclc.org/content.aspx?bookid=3107&sectionid=260925592

Access Medicine Sherris & Ryan's Medical Microbiology, 8e, 2022; Chapter 21: Bacteria – Basic Concepts https://accessmedicine.mhmedical.com.idm.oclc.org/content.aspx?bookid=3107&sectionid=260925592

ScholarRx: General Microbiology > Bacteria > Bacterial Structure and Function > Prokaryote Structure and Gram Stain https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_MIC0011

The Sanford Guide to Antimicrobial Therapy, 52e, 2022

UpToDate: Various articles; literature reviews current as of July 2024

SUMMARY: Applying Pharmacodynamics and Pharmacokinetics Principles to Antimicrobial Therapy

- Antimicrobial agents work by targeting specific elements in biochemical pathways of pathogens.
- Important determinants of success of antimicrobial therapy include:
  - proper selection of antimicrobial therapy based on their selective toxicity to the microorganism and spectrums of activity,
  - laboratory identification and susceptibility testing,
  - dosing that achieves the proper concentration at the site of action,
  - knowledge of drug penetration into the infected compartment, and
  - a dosing schedule that maximizes antimicrobial effect.
- Drug interference with physiological pathways kills (microbicidal effect) susceptible microorganisms or inhibits their growth and replication (microbiostatic effect), in which case the host immune system eradicates the pathogen.
- Some microorganisms are intrinsically resistant to certain antimicrobials – the antimicrobial cannot access the target or the pathogen lacks the specific biomolecular target.
- Acquired resistance arises through natural selection and selection pressure due to the presence of the antimicrobial agent.
  - Reduced permeability of the drug to microbe; the drug cannot reach its target
  - Active efflux of the drug out of the microbe
  - Enzymatic inactivation of the drug
  - Target alteration
  - Development of an alternative metabolic pathway
- The relationship between drug concentration and effect is modeled using concentration-effect curves depicting the concentration of a particular antibiotic needed to inhibit the growth of a particular microbe.

- The proper dose and dosing schedule are determined by integrating microbial pharmacokinetic-pharmacodynamic information, expected pharmacokinetic variability, and the minimum inhibitory concentration of the pathogen.
- The PK-PD profiles of antimicrobials that reflect the drugs' rate and extent of killing are:
  - Peak/MIC concentration-dependent effects (the higher the concentration the greater the killing rate and effect) and persistent suppression of microbial growth
  - T>MIC time-dependent effects with minimal persistent effects
  - AUC/MIC time-dependent effects with moderate to prolonged persistent effects.
- The goals of antimicrobial therapy should be clear.
  - Empiric therapy
  - Definitive therapy
  - Prophylactic therapy
  - Pre-emptive therapy
  - Suppressive therapy
  - Treatment goals and the duration of therapy should be clearly spelled out in the beginning of therapy, based on proper evidence.
- The general rule is monotherapy, except in select situations where combination therapy has been shown to be superior.
- Poor dosing strategies lead to catastrophic outcomes such as drug-resistant pathogens and untoward toxicity to the patients.
- Antimicrobial stewardship programs are coordinated practices that promote the proper use of antimicrobials, improve patient outcomes, reduce the development of microbial resistance, and decrease the spread of infections caused by multidrug-resistant organisms.
  - The goals are to:
  - reduce inappropriate use and consumption of antimicrobial agents,
  - reduce *C. difficile* infections,
  - increase adherence to antimicrobial treatment guidelines,
  - improve patient outcomes and adverse drug events, and
  - limit antimicrobial resistance.

SUMMARY: OPTIMIZING DRUG THERAPY IN THE INDIVIDUAL PATIENT

1. Use antibiotics only when a microbiologic diagnosis indicates effectiveness.
2. Empiric therapy should be tailored to the most likely pathogen(s).
3. Send appropriate cultures before starting antibiotics.
4. Select the drug(s) that penetrates body tissues to the site of infection.
5. Use narrow-spectrum antibiotics whenever possible.
6. Stop antibiotics as soon as appropriate to reduce adverse effects, such as antibiotic-associated colitis caused by *Clostridioides difficile*.
7. Be aware of the effect of the patient's renal function on the dose of antibiotic prescribed.
8. Be aware of the patient's hypersensitivity to specific antibiotics.
9. Determine whether the patient's declared hypersensitivity is correct and clinically significant.
10. Warn patients regarding certain drug reactions, such as photosensitization.

A MBLER CLASSIFICATIONS OF β-LACTAMASES
Expressed by many gram-negative bacilli

<table>
<tr><td colspan="2">Class A

ESBLs*: Extended spectrum β-lactamases typically expressed by plasmid-mediated TEM, SHV, CTX-M genes
➢ Multidrug resistance (MDR): Confer resistance to extended-spectrum cephalosporins – ceftriaxone, cefotaxime, ceftazidime – and to cefepime, cefazolin, penicillins, aztreonam
➢ Community- and hospital-acquired MDR microorganisms

Klebsiella pneumoniae Carbapenemases (KPCs): High-level multidrug resistance – ALL beta-lactams – in many gram-negative bacilli
➢ Patients with multiple comorbidities, intensive care unit stays, and prolonged hospitalizations are at highest risk of infection with KPC-expressing bacteria.
➢ High rate of mortality (40%-50%)
–ESBLs and KPCs are inhibited by all beta-lactamase inhibitors–</td><td rowspan="3">Serine beta-lactamases</td></tr>
<tr><td>Class C AmpC-type</td><td>Cephalosporinases (narrow-, extended-, and broad-spectrum) and inactivate penicillins
Carbapenems are stable against AmpC type (they remain active).
Typically, chromosome-encoded and inducible; also plasmid encoded

–Inhibited by the newer beta-lactamase inhibitors, avibactam, vaborbactam, relebactam, and durlobactam, but not the older beta-lactamase inhibitors, clavulanic acid, sulbactam, and tazobactam–</td></tr>
<tr><td>Class D</td><td>OXA: several molecular subgroups
Inactivate penicillin, oxacillin and related drugs, and 1st-gen cephalosporins</td></tr>
</table>

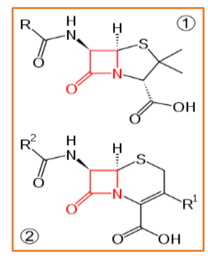
	Some OXA subtypes confer resistance to extended-spectrum cephs. Some OXA types confer resistance to carbapenems. ***–Inhibited only by avibactam beta-lactamase inhibitor–***	
Class B	Carbapenemases (classified as IMP, VIM, GIM, SPM, SIM, and New Delhi NDM-1) Chromosomally encoded; both naturally occurring and acquired ***–Not inhibited by beta-lactamase inhibitors–***	Metallo-beta-lactamases

FYI: Structural features: Identity of β-lactamase by amino acid sequence

1. Serine proteases Class A, C, and D: these enzymes all have a serine residue in the active site
2. Metalloproteinases Class B: zinc ion(s) in the catalytic site required for hydrolysis activity
3. Carbapenemases: Organized based on amino acid homology in the Ambler molecular classification system. Class A, C, and D beta-lactamases all share a serine residue in the active site, while Class B enzymes require the presence of zinc for activity (and hence are referred to as metallo-beta-lactamases).

SUMMARY: BETA-LACTAM ANTIBIOTICS, OTHER CELL WALL, MEMBRANE INHIBITORS

PART 1



- Structure-activity relationship:
  - β-lactam ring (D-alanyl-D-alanine analog)
  - A secondary amino group (RNH–) to which substituents (side chains) are attached
- Pharmacokinetics, Class properties:
  - A: Absorption from the GI tract is dependent on the drugs' acid stability; drugs that are unstable in acid (acid labile) have poor bioavailability; Parenteral administration (IM, IV): Rapid, predictable plasma concentrations of drug
  - D: Widely distributed in body fluids and tissues; significant amounts appear in liver, bile, kidney, semen, joint fluid, lymph, and intestine

    Most beta-lactams poorly penetrate phagocytic cells, as well as the prostate, the eye, and the cerebral spinal fluid (CSF), which have tight junctions.

    ⇒ *Clinical correlate: When meninges are inflamed, as in bacterial meningitis, drug concentrations in CSF increase.*
  - Elimination: Primarily renal excretion; a few beta-lactams have some degree of hepatic elimination
- PK-PD Profile, T>M: Time-dependent (concentration-independent) and minimal persistent effects
- Bactericidal

Exception: Bacteriostatic during the stationary phase

- Mechanism **leading to cell death by osmotic rupture**
  1) Beta-lactams covalently bind to transpeptidases (PBP) →
  2) Block with the peptide cross-linking →
  3) Interferes with cell wall synthesis →
  4) Triggers the degradative action of bacterial autolysins →
  5) Destruction of the existing cell wall (autolysins participate in the normal remodeling of the cell wall) and new cross-links fail to form →
  6) Osmotic rupture of bacterial cell
  7) Non-lytic mechanisms causing collapse of membrane potential also appear be involved.
- Resistance mechanisms
  1) Production of β-lactamases → inactivation of drug
  2) Modification of penicillin binding protein (PBP) (the drug target) → low affinity binding of drug

  Penicillin-resistant staphylococci, pneumococci, and enterococci

  3) Impaired penetration of the impermeable outer membrane of gram-negative bacteria → the drug does not reach the target PBP

  – Gram-negative bacteria form aqueous channels in the outer membrane, called ***porins***, which allow small polar molecules, including some antibiotics, to penetrate the outer membrane.

  – Antibiotic efflux by gram-negative bacteria → drug is transported out of periplasmic space back across cell wall outer membrane

  4) Obligate intracellular bacteria (beta-lactams do not penetrate into phagocytes)
  5) Bacteria that lack peptidoglycan cell wall (*Mycoplasma*)
- Spectrum of action: Varies from narrow to extended-spectrum to broad-spectrum. Please see the notes handout or textbooks for details.
- Adverse effects:
  – Allergic reactions: include urticaria, severe pruritus, fever, joint swelling, hemolytic anemia, nephritis, and anaphylaxis.
  – Complete cross-reactivity of penicillins
  – Nafcillin is associated with neutropenia.
  – Ampicillin frequently causes maculopapular skin rash that does not appear to be an allergic reaction.

- Nausea, diarrhea
- Secondary infections

Secondary infections

- *C. difficile*-associated diarrhea, pseudomembranous colitis, toxic megacolon
- Yeast infections

- β-lactamases and β-lactamase Inhibitors:

Knowledge of resistance mechanisms is very important for all members of the health care team. Microbial resistance is a worldwide, potentially deadly, problem.

- Beta-lactamase inhibitors have minimal antibacterial activity. The antibacterial spectrum is determined by the beta-lactam antibiotic, not the beta-lactamase inhibitor.
  - Clavulanate, sulbactam, tazobactam: Suicide inhibitors of beta-lactamases
  - Avibactam, vaborbactam, relebactam: high-affinity serine protease inhibitors; competitive inhibitors
  - They inhibit the activity of a number of plasmid-mediated beta-lactamases
  - Ambler class A TEM β-lactamases of staphylococci, *H. influenzae, N. gonorrhoeae, E. coli*, *Klebsiella, Salmonella, Shigella*; *Bacteroides*, and *Moraxella*
  - Avibactam and Relebactam inhibits chromosomally mediated AmpC beta-lactamases.
  - Combinations of beta-lactamase inhibitors with ampicillin, amoxicillin, ticarcillin, piperacillin, ceftolozane, or ceftazidime results in antibiotics with an enhanced spectrum of activity against many, but not all, organisms containing plasmid-mediated beta-lactamases.
- One exception: **Sulbactam-Durlobactam**

Sulbactam is bactericidal against *A. baumannii*, a gram-negative coccobacillus associated with health care-associated infections in the ICU and traumatic community-acquired infections. Sulbactam binds *A. baumannii* transpeptidases PBP1 and PBP3.

Durlobactam inhibits beta-lactamases expressed by the microorganism.

SUMMARY: BETA-LACTAM ANTIBIOTICS, OTHER CELL WALL, MEMBRANE INHIBITORS PART 2

Pharmacokinetics

- The pharmacokinetic properties of the cephalosporins, carbapenems, and aztreonam are typical of the beta-lactams class.
- Ceftriaxone is excreted via the biliary system and renally.
- Vancomycin and daptomycin are also renally excreted as unchanged drug.

- Monitor renal function. Dose adjustment is required in patients with renal impairment.

Mechanisms of action

- The cell-wall inhibitors kill bacterial cells only when they are actively growing and synthesizing the cell wall.
- Like all beta-lactams, the cephalosporins, carbapenems, and aztreonam covalently bind the active site of PBP, which halts cell wall synthesis and leads to cell death; autolysins are involved. The spectrums of action vary with the individual drugs. Aztreonam is effective only against gram-negative, aerobic bacteria.
- Cefiderocol is a siderophore that complexes extracellular ferric iron and is actively transported across the outer member of aerobic gram-negative bacteria by iron transport systems.
- Vancomycin binds to D-ala-D-ala of the pentapeptide on the lipid carrier-NAM-pentapeptide precursor, which prevents transglycosylation and polymerization of the glycopeptide subunits.
- Daptomycin binds to the cytoplasmic membrane by inserting its lipid tail in a calcium-dependent process. A complex is formed in the bacterial plasma membrane, which causes efflux of potassium, that leads to the arrest of DNA, RNA, and protein synthesis resulting in cell death.
- Vancomycin and daptomycin are active only against gram-positive aerobic and anaerobic bacteria.

Mechanisms of resistance

- Cephalosporins, carbapenem, and aztreonam are inactivated by ESBLs and KPC carbapenemases. Combining with beta-lactamases restores the activity.
- Beta-lactams are ineffective against organisms expressing altered PBPs.
- Beta-lactams are also ineffective against obligate intracellular pathogens and those that do not have a peptidoglycan wall.
- Aztreonam binds poorly to PBP sites of the aerobic gram-positive and anaerobic bacteria and is, therefore, ineffective against these organisms.
- Vancomycin resistance occurs in staphylococci that produce thickened cell wall and enterococci that express modified D-alanyl-D-lactate. Staphylococci acquire D-ala-D-lac by plasmid transfer.
- Daptomycin resistance among staphylococci and enterococci is occurs infrequently. The mechanism has not been fully characterized.

Therapeutic uses

- As with all antimicrobial therapy, eradication of the infectious organisms at the site of infection is the goal of treatment.

- Therapeutic selection (empiric and definitive therapy) is based on the infectious pathogen, location and severity of infection, activity of antimicrobial against the specific pathogen, potential adverse effects, patient factors and contraindications, and cost.
- Prophylactic antibiotic selection to reduce surgical site infections is based on the likely causative pathogens.
- Most recommendations in infectious disease guidelines are based either on expert opinion or evidence-based medicine.

Adverse effects

- Some adverse effects are unique to certain antibiotics.
- *C. difficile* and *Candida* secondary infections can occur.
- Cephalosporins can elicit hypersensitivity reactions, including anaphylaxis, fever, skin rashes, nephritis, granulocytopenia, and hemolytic anemia. Cross-reactivity, although frequency is low, can occur in patients with penicillin allergy.
- Ceftriaxone can cause biliary colic and cholecystitis; hyperbilirubinemia and kernicterus risk in neonates; precipitation with calcium; ceftriaxone-calcium precipitates associated with fatal lung / kidney damage in newborns.
- Cefotetan may cause hypothrombinemia and bleeding disorder.
- Cefiderocol is useful against MDR aerobic GNBs.
- Carbapenems may cause seizures; GI effects are common; bleeding and leukopenia have been reported.
- Aztreonam is generally well tolerated.
- Vancomycin can cause nonimmunogenic flushing and pruritis of the face, neck, and trunk, which is called non-immune anaphylactoid infusion-related reaction (red man syndrome); hypotension, dyspnea, muscle spasms may also occur. High concentrations are associated with nephrotoxicity and ototoxicity. The risk increases with concomitant use of other drugs that cause similar toxicities. Intramuscular injection of vancomycin is contraindicated (tissue necrosis).
- Daptomycin's primary toxicity is myopathy; CK must be monitored. Eosinophilic pneumonia and peripheral neuropathy have been reported. The drug should be discontinued immediately if signs of any of these adverse effects occur. Consider avoiding concomitant use with other drugs that cause similar adverse effects.